Tile Drainage: Research Frontier

1 Abstract

Agricultural subsurface (tile) drainage underpins crop production on poorly drained and irrigated land, yet the same conduits that make these soils farmable also short-circuit the landscape's natural buffering, delivering nitrate, phosphorus, salts, and a widening list of constituents to ditches, streams, and coastal waters. Decades of field experiments, monitoring, and modelling have produced a detailed literature on drained-system behaviour and on the practices—nutrient management, controlled drainage, cover crops, edge-of-field treatment, salinity reclamation—that might reconcile productivity with water quality (Randall and Mulla 2001). This review is the research-frontier component of a larger synthesis, framed deliberately around what remains uncertain, untested, or unstudied rather than around settled results. Its recurring meta-finding: ideas have outrun the data able to test them— headline results rest disproportionately on short records, single sites, and narrow agro-ecological windows, and progress in every domain is bounded by the length, density, replication, and consistency of the underlying observations. We proceed from enabling to domain frontiers, then to consequences and integration. The enabling foundations are an observational base of longer, denser, and harmonised monitoring; process-based drainage and water-quality models whose credibility is limited less by their formulation than by the narrow envelope in which they have been validated; and the coupled problems of mapping the largely unmapped drainage network and scaling instrumented-field knowledge to the watersheds where water-quality targets are set. On this footing we treat the domain frontiers of nitrogen, phosphorus, and cropping-system losses; active water-table management and drainage-system redesign; edge-of-field treatment such as bioreactors, wetlands, and phosphorus-removal structures; and salinity reclamation, drainage-water reuse, and disposal. Because removing a constituent from one pathway can shift it to another and altering the field water regime can trigger slow biogeochemical and pedological change, we then consolidate the unintended consequences and multi-objective tradeoffs—greenhouse-gas coupling, pollution swapping, emerging contaminants, trace-element legacies, and drainage-induced soil evolution—that nitrate- and yield-centred accounting has overlooked. Finally, we draw the integrative strands together: no single practice or model is a silver bullet, and decisive advances will come from coupling mechanistic understanding with spatial targeting, economics, policy, and shared decision-support infrastructure. The unifying message is that the next decade of progress will be secured less by new equations or novel practices than by sustained, well-instrumented, openly reported, and geographically diverse field campaigns, coordinated data standards, and interdisciplinary integration tying mechanism to tools and tools to adoption.

2 Introduction

Agricultural subsurface (tile) drainage is one of the foundational technologies of modern crop production: by lowering the water table on poorly drained land it secures trafficability, root aeration,

and yield across vast areas of the temperate and irrigated world. Yet the same engineered conduits that make these soils farmable also short-circuit the landscape's natural buffering, delivering nitrate, phosphorus, salts, and an expanding list of other constituents rapidly to ditches, streams, and ultimately to coastal waters. Decades of field experiments, monitoring campaigns, and modelling have produced a large and detailed literature on how drained systems behave and on the practices—nutrient management, controlled drainage, cover crops, edge-of-field treatment, salinity reclamation—that might reconcile productivity with water quality. The distinctive feature of that literature, however, is that its most secure and most frequently repeated statements are not about what is known but about what remains unknown. This review is organised around that observation.

Read across the field, the recurring signal is a precondition rather than a conclusion. Headline results are drawn disproportionately from short records, single sites, and narrow agro-ecological windows, and their authors caution explicitly against premature generalisation. The blunt appeals are pervasive—that "more research is needed" (SHALIT et al. 1995), that "the complexity of this subject requires more research" (Malone, Nolan, et al. 2014), that "further research in this area is needed" (Woo, Ji, and Song 2023), and that "further evaluations are warranted before any concrete conclusions could be drawn" (GUPTA et al. 1993)—and they mark genuine gaps in evidence rather than rhetorical closings. Rogger et al. (2017) pitch the same concern at the level of the whole discipline, calling for complex-systems thinking that links processes across time scales, long-term experiments on physical–chemical–biological interactions, attention to connectivity and pattern across spatial scales, and multidisciplinary consortia that share data and terminology under known boundary conditions. The binding constraint that emerges from these appeals is not a single missing experiment but a coupled deficit: coordinated, long-term, spatially replicated, and harmonised data on the one hand, and better process representation in models on the other, must advance together, because neither longer records nor more elaborate models resolve the field's open questions alone.

This paper is the research-frontier component of a larger review of tile drainage, and it is deliberately framed around uncertainty. Where the companion component consolidates settled knowledge, our purpose here is to gather, structure, and sharpen the open questions—the priorities, the hypotheses that remain to be tested, and the methods, instrumentation, and datasets still needed to test them. We therefore do not attempt to resolve any of the gaps we identify; we set them up, tracing how each is posed in the primary literature and how the strands interlock. The aim is a map of where the field should go next, in which the confidence of a claim about what is *not* yet understood is treated as a finding in its own right.

The synthesis proceeds from enabling frontiers to domain frontiers and then to their consequences and integration. We begin with the cross-cutting foundations on which every later domain depends: the observational foundation of longer, denser, and harmonised monitoring; the process-based drainage and water-quality models whose credibility is limited less by their formulation than by the narrowness of the conditions under which they have been tested; and the linked problems of mapping the largely unmapped physical drainage network and scaling instrumented-field knowledge to the watersheds where water-quality targets are actually set. On this footing we turn to the domain frontiers—managing nitrogen, phosphorus, and cropping-system losses; active water-table management and the re-evaluation of drainage-system design; edge-of-field treatment technologies such as denitrifying bioreactors, wetlands, and phosphorus-removal structures; and salinity reclamation, drainage-water reuse, and disposal in arid and irrigated systems. Because removing a constituent from one pathway can shift it to another and altering the field water regime can set slow biogeochemical and pedological change in motion, we then consolidate the unintended consequences and multi-objective tradeoffs—greenhouse-gas coupling, pollution swapping, emerging contaminants, trace-element legacies, and drainage-induced soil evolution—that a nitrate- and yield-centred accounting has historically overlooked. Finally, we draw the integrative strands to-

gether, because translating frontier science into outcomes depends on how practices are bundled and spatially targeted, how they perform under a changing climate, and whether they are economically and institutionally adoptable.

A single thread runs through all of these: no individual practice or model is a silver bullet, and the decisive advances will come from coupling mechanistic understanding with spatial targeting, economics, policy, and shared decision-support infrastructure. Underlying that ambition is a data imperative—better, longer, and more spatially representative datasets—that recurs as the foundational requirement for validating models, resolving tradeoffs, and delivering site-specific, climate-resilient guidance (Z. Ma et al. 2025; SKAGGS, BREVE, and GILLIAM 1994; Deng et al. 2011). The sections that follow elaborate that agenda frontier by frontier.

3 The Observational Foundation: Longer, Denser, and Harmonised Monitoring

Across the drainage literature, the field's most secure statements are about the limits of its own evidence. Headline results are drawn disproportionately from short records, single sites, and narrow agro-ecological windows, and their authors caution explicitly against premature generalisation. The recurring signal is therefore not a settled conclusion but a precondition: records long enough, dense enough, spatially replicated enough, and consistent enough to separate a management signal from the natural variability of weather, hydrology, and slow soil change. Rogger et al. (2017) pitch this at the level of the whole discipline, calling for complex-systems thinking that links processes across time scales, long-term experiments on physical–chemical–biological process interactions, attention to connectivity and patterns across spatial scales, typologies and meta-analyses that generalise findings, and multidisciplinary consortia sharing data and terminology under known boundary conditions. The sheer frequency of blunt appeals—that "more research is needed" (SHALIT et al. 1995), that "the complexity of this subject requires more research" (Malone, Nolan, et al. 2014), that "further research in this area is needed" (Woo, Ji, and Song 2023), and that "further evaluations are warranted before any concrete conclusions could be drawn" (GUPTA et al. 1993)—marks genuine gaps in evidence rather than rhetorical closings. Because every later domain in this review inherits these limits, the observational foundation is treated here as a frontier in its own right.

3.1 Extending the temporal record

The single most widely voiced need is for records long enough to disentangle treatment effects from climatic noise and from slow, directional change. Clifford et al. (2023) emphasise truly long-term experiments, and BUHLER et al. (1993) caution that a lag of several years may precede any objective evaluation of an altered practice. Minimum horizons are proposed—at least five years, with continued observation as systems are extended in length or depth over time (R. Ali et al. 2013), and lengthening national time series expected to sharpen analysis as records grow (A. Ali, Van Leeuwen, and Koopmans 2001). Many studies attach conclusions to explicit requirements for further years: to quantify a denitrification wall's effectiveness and its long-term cost per unit nitrate removed against natural variability (Jaynes, Kaspar, et al. 2008), to establish how long treatment benefits persist (Dalhem et al. 2019; Osterholm, Åström, and Sundström 2005), and to test the replicability of results in time and confirm that observed differences endure under comparable conditions (ROLLIN, BROUGHTON, and BOLDUC 1987; Reinhart et al. 2020; I. Rijal et al. 2012; Giménez-Valero et al. 2025), all echoing the disciplinary case for long-term monitoring at multiple scales (SPALING 1995).

Nutrient and hydrological studies converge on the same horizon. Year-round measurement of flow and concentration for at least several years on new sites is deemed necessary to set management priorities (Kladivko, Brown, and J. Baker 2001), and long-term records are singled out to close water and nutrient balances and study legacy pools (Sharpley et al. 2019) and to disentangle land-management from climate-change effects, with the corollary that policy evaluation demands patience (Bauwe et al. 2020; Kelley et al. 2017; Gillingham and Thorrold 2000; Lemke et al. 2011). Smoothing year-to-year variability requires evaluating both phases of a rotation (Randall, Vetsch, and Huffman 2003; Randall and Vetsch 2005), assessing responses on an annualised, multi-year basis (Waring, Pederson, et al. 2024; Pittelkow et al. 2017), and freeing drainage-water-management yield effects from confounding (Cooke and S. Verma 2012); only long-term records across a range of climatic conditions, gathering plant, soil, and water measurements together, yield appropriate guidelines (RANDALL and IRAGAVARAPU 1995). Data spanning at least a normal, a wet, and a dry year are needed to resolve losses and transport routes (Salm et al. 2012), and as long a record as possible is required for trend estimation because water quality may lag the management-driven change (Ulén, Bechmann, et al. 2007). Long records are equally the only way to detect slow processes—soil organic-carbon recovery (Kumar, Nakajima, Mbonimpa, et al. 2014), discharge-driven organic-matter export (Manninen, Kanerva, et al. 2023), and whether drainage-induced soil evolution continues or stabilises over decades (Frison et al. 2009; Frankl et al. 2023). Record length alone is insufficient: robust seasonal conclusions require year-round coverage, especially in the non-growing season (Klaiber, Kramer, and E. Young 2020; Van Esbroeck et al. 2016), high-frequency records without gaps and instruments checked as often as possible (Hodaj et al. 2017), and monitoring timed to infrastructure interventions such as drain repair (Wesström, Ulén, et al. 2015).

3.2 Broadening spatial and geographic scope

A second frontier is testing whether site-specific findings hold across other soils, climates, crops, and management regimes. Authors repeatedly defer broad adoption until similar data are collected from further sites and years (Menegaz et al. 2025; Zimmerman and Kaleita 2017; Terrado, Tauler, and Bennett 2015), from additional agro-ecological zones (A. Ali, Van Leeuwen, and Koopmans 2001), from other settings for practices such as the two-stage ditch (Hodaj et al. 2017), and from adverse or compacted conditions where a technology's merit is untested (Bigelow et al. 2001). The most explicit prescription is that controlled-drainage and subirrigation studies be conducted outside the U.S. Midwest and Canada across diverse climates, crops, and soils, and— crucially for generalisation—that site characterisation (slope, particle-size distribution by horizon, hydraulic conductivity, drainage class, and moisture) be reported with measures of variability (N. Singh et al. 2022). Repetition at larger scales and on contrasting soils is stressed as a key modulating factor (Bilotta et al. 2008; Goeken, X. Zhou, and Helmers 2015), including for process studies such as earthworm activity on soils of differing texture and rainfall (Nuutinen, Pöyhönen, et al. 2001; Nuutinen, Butt, and Jauhiainen 2011). Climate-focused transferability demands high-resolution scenarios and multi-model uncertainty analysis at additional sites (Rasmussen et al. 2018; Jeantet, Thirel, et al. 2022; Jin et al. 2008), and testing across regions, soils, and climates recurs throughout as the precondition for localised design guidance (S. Singh et al. 2020; Malota, Taguta, et al. 2024; Salazar, Wesström, and Joel 2008; Holbak et al. 2021). At the level of synthesis, robust conclusions will require geographically diverse and methodologically harmonised studies so that meta-analyses can disentangle management effects (R. Wu et al. 2025).

3.3 Monitoring infrastructure, instrumentation, and data tools

A distinct set of needs concerns the apparatus that makes long-term data collection feasible and reliable. Kanwar, Bjorneberg, and D. Baker (1999) describe how automatic samplers collect discrete alongside continuous samples, how data loggers can be extended to soil temperature and moisture, and how relays or pressure switches could detect sump-pump operation, advising that equipment-to-logger distances be minimised, data downloaded automatically to reduce loss, and annual maintenance—cleaning check valves and calibrating flowmeters—be performed. Diagnostic methods are themselves research targets: reducing pressure variations by doubling equipment and injecting smoke from both ends of a drain section could enable work with longer drain lines, though transport pathways and observed clustering remain unresolved (Petersen, Nielsen, and S. Hansen 2012), and selected pipes, envelopes, perforations, and surrounding soil should be observed with permanent monitoring of the sampling process to confirm favourable results are maintained (Martínez-Beltrán, González-Casillas, and Namuche-Vargas 2007). Closing the water budget requires high-frequency discharge, runoff, and seepage measurement, process measurements of microbial activity, denitrification, and redox, and a publicly available dataset (Frankenberger et al. 2024). Software and data infrastructure are an increasingly prominent part of this frontier: readers are directed to evolving data products such as the National Elevation Dataset and USGS resources (Shedekar and Brown 2018), and a series of drainage-network configuration and sizing tools describe themselves as under continual development and openly invite testing with additional datasets (Falasy and Cooke 2025a; Falasy and Cooke 2025b; Falasy and Cooke 2024).

3.4 Tracers, isotopes, and source apportionment

Measurement and tracing techniques are repeatedly identified as the means to resolve pathways that bulk flow-and-concentration records leave ambiguous. Williams, Penn, and McAfee (2022) call for higher-frequency (sub-daily to daily) measurements, further work on groundwater dynamics in dissolved-P transport, and more comprehensive source-apportionment methods. Water- and nitrate-isotope approaches are advanced to quantify processes directly: coupling water and nitrate oxygen isotopes and quantifying the $\delta^{18}O$ of nitrification-derived nitrate to constrain evaporative enrichment (Y. Hu et al. 2024), monitoring nitrate concentrations and isotopic compositions with multiple tracers across topsoil, drains, and ditches to characterise the soil contribution to river nitrate (Briand et al. 2017), and boron isotopes to strengthen contaminant tracing around feedlots and fields (Komor 1997). High-frequency tracer measurements are urged to quantify water ages and represent both celerities and velocities (Williams and McAfee 2021), alongside event sampling and bulk-soil-water isotopes to constrain young-water fractions (Dimitrova-Petrova et al. 2020), high-resolution specific conductance for end-member mixing (Kronholm and Capel 2015; Schilling, Gassman, et al. 2019), and conservative-tracer or water-isotope studies to partition quick flow and baseflow (Frederiksen et al. 2023). Explicit pathway partitioning— reporting surface-inlet presence and splitting water and nutrient loads by pathway via before-after control-impact designs, hydrograph separation, or tracers (Flores, Bailey, and Harmel 2021), evaluating overland-flow versus tile-flow contributions (Kokulan et al. 2019), and verifying groundwater flow paths and plant water sources with stable isotopes (Islam et al. 2022; Yli-Halla et al. 2022)—recurs throughout. The role of groundwater outflow in nutrient loading is flagged as under-quantified and in need of dedicated measurement and modelling (Turunen, Warsta, et al. 2013; Jokinen et al. 2024; Barkle et al. 2021), and finer temporal analyses of the flow–concentration relationship, including detailed storm-hydrograph separation, are needed to test event-scale hypotheses (Lagzdins et al. 2015; Danalatos et al. 2022; Kladivko, Frankenberger, et al. 2004).

3.5 Standardisation, harmonisation, and comparability

For accumulating data to support synthesis, methods and reporting must be harmonised. The value of future meta-analyses is tied directly to methodologically harmonised, comprehensive, long-term monitoring of the full suite of environmental impacts (R. Wu et al. 2025). T. Zhang, Z. Zheng, et al. (2018) set out the most detailed programme: improving the quality and comparability of existing indicators through harmonised data collection, monitoring, interpretation, and reporting; developing new indicators for emerging concerns while ensuring scientific credibility, sensitivity to change, and ease of interpretation; establishing standardised evaluation procedures including thresholds and reference points; recognising data gaps and improving upscaling and uncertainty assessment; and incorporating earth-observation data. These priorities complement the call for typologies, meta-analyses, and shared data and terminology across research consortia (Rogger et al. 2017). At the study level, meta-analytic comparability requires complete hydrologic description of each plot, standardised analytical methods, adequate monitoring periods, and full reporting of concentrations with descriptive statistics (Ghirardini and Verlicchi 2019), and soil biological and soil-health indicators must be sampled at consistent times under similar moisture and temperature conditions to be comparable across sites (Sherbine et al. 2023; N. Li et al. 2023).

3.6 Experimental design for detectable effects

Longer, standardised records are useful only if the design can resolve the effect sought. Schilling, Jones, and Seeman (2013) specify the ingredients for paired-basin detection: establishing a suitable correlation among paired basins across climates, restricting paired sampling to the March–July high-flow period, delaying post-treatment monitoring to enhance the chance of detecting change, and adopting practices capable of achieving at least the minimum detectable change. Effects should be assessed simultaneously at multiple scales, from field to watershed (Sunohara, Gottschall, Craiovan, et al. 2016), paired with within-field and within-plot studies to elucidate the mechanistic processes of export and additional field-scale studies within high-export sub-watersheds (Irvine et al. 2019). Larger, more representative trial areas are sought to test effective-yield formulas (Xin, Dan, et al. 2016b) and to refine tracer-based (^{15}N) loss observation (Xiong et al. 2025), and mesocosms are proposed to quantify nutrient-loss responses to management change under controlled conditions (Loper et al. 2024). Because field-scale drainage trials are typically unreplicated, several authors treat the analysis itself as an open question: whether beta distributions, time series, or new methods best analyse unreplicated paired experiments (Pagliari, Ranaivoson, and Strock 2021), and whether directed hypothesis tests should be favoured over replicate spatial analyses (Andrus et al. 2014). Where replication is feasible, replicated field experiments, matched instrumentation, and improved sampling designs are recommended to support statistical inference (Okuda et al. 2020; Wamono et al. 2016; Davis et al. 2019; Jacquemin et al. 2020). Together these design refinements define the bridge between a lengthening, harmonised observational record and the detectable, transferable conclusions the rest of the field depends upon.

4 Advancing Process-Based Drainage and Water-Quality Models

Process-based drainage and water-quality models—DRAINMOD and its derivatives, SWAT, APEX, RZWQM2, HYDRUS, MIKE SHE, and numerous salinity codes—are expected to carry much of the predictive burden as field trials cannot span every combination of practice, soil, and climate. Yet across the literature their credibility is constrained less by their formulation than by the narrowness of the conditions under which they have been tested: models are typically calibrated at

a single site, over a handful of seasons, for one soil–climate–management combination, and their authors are nearly unanimous in flagging broader empirical validation, missing process data, and weak algorithmic components as the outstanding priorities (C. Li et al. 2006; Z. Qi, R. Singh, et al. 2015; Youssef, Skaggs, et al. 2006; Radcliffe et al. 2015; SKAGGS, BREVE, and GILLIAM 1994). This section gathers the model-algorithm and model-structure gaps that recur throughout the field.

4.1 Broadening the empirical basis for validation

The most frequently articulated need is validation data spanning a wider range of conditions than any single study can supply. Youssef, Skaggs, et al. (2006) describe rigorous field testing across differing management, soils, and climates as a need common to nearly all simulation models, and warn that assumptions must be re-verified whenever a code is transferred to a new area (Kelleners, Kamra, and Jhorar 2000); similar calls for broader-condition testing, sensitivity analysis, and parameter refinement recur for pesticide, salt, and flow models (Branger et al. 2009; Hirekhan, S. Gupta, and Mishra 2007; X. Yang 2008; Xian et al. 2017). Models are repeatedly reported as requiring testing over longer periods and additional soil types (Salazar, Wesström, Youssef, et al. 2009; W. Li et al. 2026), and their effectiveness under new controlled-drainage or erosion conditions is judged critically needed before higher credibility can be claimed (Negm, Youssef, and Jaynes 2017; Turunen, Turtola, et al. 2020).

A recurring, more ambitious hypothesis is that empirical coefficients could be replaced with functions of measurable site attributes. C. Li et al. (2006) commit to testing across climate zones, soils, and management while anticipating that coefficient constants might become functions of soil texture, converting site-specific tuning into transferable structure; Chelil, Henine, et al. (2022) cast the same concern as spatial robustness, arguing that additional fields are needed to confirm that parameters such as nitrate-transfer velocities are transferable rather than calibration artefacts. Verification at independent sites is treated as a precondition, not optional confirmation (Azevedo et al. 1997; Z. Qi, R. Singh, et al. 2015; Kalita et al. 1998; Sadhukhan, Z. Qi, T. Zhang, Tan, and L. Ma 2019), with inconsistencies producing conflicting parameter values resolved first (Dar and J. Singh 2022) and empirical codes transferred only through re-parameterisation with local soil, geology, crop, and climate data, or region-specific coefficients where reference values exist (Azevedo et al. 1997; Fath-Allah 2025; S. Rijal, X. Zhang, and X. Jia 2013). The open question is how much local data such transfer requires. Extending validation across the climatic and design envelope is closely related: humid-region models need testing under semi-arid regimes and after prolonged dry spells (Prasher et al. 1996), longer monitoring records give firmer conclusions (Ebrahimian, Parsinejad, et al. 2011; Grazhdani, Jacquin, and Sulce 1996), and design geometry demands experiments at spacings wider than 30 m and regional answers on drainage-efficiency loss and design storms (Mathew, Panda, and Nair 2001; Wiskow and Ploeg 2003; MULLER 1988). Long simulations of 50–100 years are argued necessary to judge whether a design meets its criteria (Mendez et al. 2004; X. Wang, Mosley, et al. 2006; Ale et al. 2009; Häggblom et al. 2019). Portability, in short, is a hypothesis to be tested, not a property to be presumed (Hornbuckle, Christen, and Faulkner 2012; Shipitalo, Malone, et al. 2016; Yousfi, Mechergui, and Ritzema 2014).

4.2 Field and process data still needed to constrain models

Beneath the demand for broader validation lies a scarcity of process-level measurements. Nitrogen dynamics emerge as the least well-constrained component: David, Del Grosso, et al. (2009) argue that systematic denitrification measurements are required to test and refine models, their ideal being a multi-year, tile-drained corn–soybean study measuring every major N flux, with a pragmatic

fallback of a few key fluxes plus denitrification. Field- and laboratory-grounded parameterisation, denitrification foremost, is likewise judged necessary (Salazar, Wesström, Youssef, et al. 2009), together with routine evaluation and reporting of soil-N fluxes in SWAT (Ikenberry et al. 2017), long-term carbon–nitrogen estimates to reconcile plot- and subbasin-scale discrepancies (Singer et al. 2011), and resolution of the lag between practice implementation and measurable reduction (Law, Long, et al. 2022; Tao, N. Li, et al. 2021). High-frequency process measurements of microbial activity, denitrification, soil-P pools, and redox are urged for drainage water management, alongside closure of water and nutrient balances (Frankenberger et al. 2024; Sharpley et al. 2019).

Phosphorus is similarly data-limited, and its dominant pathway—preferential flow—is the hardest to observe. Many phosphorus-index factors need investigation of their effect on field-to-stream transfer, and index rankings must be shown to reflect actual transfer risk (Bechmann, Stålnacke, and Kvaerno 2007; Beauchemin et al. 2003). Confirming P-movement trends requires quantitative tracer measurement of preferential flow (Salazar, Wesström, and Joel 2011; Tiemeyer, Kahle, and Lennartz 2009), structural deficiencies around surface runoff and tile P transport need repair (Apostel et al. 2021), and accurate capture of flashy export demands automated samplers, real-time sensors, and high-resolution topographic survey of ditch sediment before and after storms (Dialameh and Ghane 2022; Warsta et al. 2014). Supporting empirical work includes further testing of $CaCl_2$-P as a dissolved-reactive-P estimator (McDowell and Sharpley 2001), regression models linking drainage parameters to P losses (Morrison, Madramootoo, and Chikhaoui 2013), tests using actual effluent under fluctuating flow (Sanford and Larson 2016), and long-term monitoring of P sources (X. Jia, DeSutter, et al. 2012). Sampling design is itself a constraint: weekly sampling may be needed to capture nitrate spikes before they pass below suction-cup samplers (Zotarelli et al. 2007), more frequent early-season and summer-event sampling is advised (Kalita et al. 1998; Tiemeyer, Kahle, and Lennartz 2009), and denser sampling geometry sharpens semi-variograms and optimal sampling density (GALLICHAND, MARCOTTE, PRASHER, and BROUGHTON 1992; Upadhyay et al. 2019).

4.3 Preferential and macropore flow representation

The representation of preferential flow is perhaps the most frequently cited frontier, the recurring theme being that its contribution to drains must be characterised as a function of soil type, water content, and practice before models extrapolate. Gaillot et al. (2023) and Fox, Malone, et al. (2004) call for mechanistic characterisation of macropore–drain connectivity and of the "express fraction" under varying saturation above the drain, and Askar, Youssef, et al. (2020) stress that rigorous testing of DRAINMOD's empirical macropore approach requires water-table-depth measurements alongside drainage and runoff, direct macropore-flow inputs, and crack geometry and dynamics. Process improvements span better matrix–macropore partitioning calibrated against sub-hourly drainage in RZWQM2 (Z. Li, Xian, et al. 2023), dynamic connectivity or effective macroporosity at watershed scale (Poon, Whalen, and Michaud 2021), micropore diffusion and site-specific conductivities in dual-permeability descriptions (Villholth, Jensen, and Fredericia 1998; Villholth and Jensen 1998), hysteresis and biopore-flow initiation (Holbak et al. 2021), manure-driven macropore development (S. Zhao et al. 2001), crack re-establishment in EPIC (Z. Wang, T. Zhang, Tan, Xue, et al. 2022), and calibrated macropore conductivity in three-dimensional dual-permeability models (Varvaris et al. 2018). Direct pathway identification is repeatedly urged—ground-penetrating radar and tracers with explicit flowpath representation (Tähtikarhu et al. 2025; H. Salo, Warsta, et al. 2017), dye-staining of routes reaching tile lines (Nielsen, Petersen, and S. Hansen 2015), and near-surface pathway characterisation (Frey, Rudolph, and Conant 2012; Allaire et al. 2011). Central quantitative questions remain the fraction of phosphorus and solutes delivered via preferential ver-

sus matrix flow (K. King, Williams, et al. 2015; Williams, K. King, Ford, et al. 2016), the range of macropore flows across soils and management (D. Reid, Ball, and T. Zhang 2012), and by-pass flow through cracks for full-year application (LESAFFRE and ZIMMER 1988; BACKLUND et al. 1995; Ruark, Brouder, and Turco 2009). Runoff simulation under crusting, cracking, and preferential flow (W. Smith et al. 2020), explicit representation of actual drain configuration (Stamm et al. 2002), temporal variability in hydrological-pathway activation for cracking clays (Kokulan et al. 2019), and seasonal, time-dependent shrink–swell and hydraulic-conductivity measurement (Boivin et al. 2006; Chelil, Oubanas, et al. 2022; Jeantet, Henine, et al. 2021) complete this agenda.

4.4 Coupled hydraulics, seepage, and soil heterogeneity

A recurrent structural gap is the absence of models coupling soil pore-water flow with surface flow inside drains and ditches. Xin, Dan, et al. (2016a) call for coupling drain surface water with pore-water flow to represent the time-varying head around the drain, and Nousiainen et al. (2015) propose simulating in-pipe flow to capture pressurisation and clogging, adding open-ditch networks. Pressurised-flow dynamics are a specific frontier: coupling pipes to the water table could introduce a validated temporary-retention function beyond design capacity (Henine, Nédélec, Augeard, et al. 2010; Henine, Nédélec, and Ribstein 2014), with pressurised periods analysed separately from pressure-transducer data (Saadat, L. Bowling, Frankenberger, and Kladivko 2018). Entrance resistance, near-drain pressure heads, and rapid trench-backfill versus slower inter-trench transport remain unresolved for heterogeneous systems (Messing and Wesström 2006; Rimidis and Dierickx 2003), and model dimensionality is itself a question—two- or three-dimensional treatment of sloping fields, collectors, and regional management (Alakukku et al. 2010; FIPPS and SKAGGS 1989; Carlier, Kao, and Ginzburg 2007; Mohanty, Bowman, Hendrickx, Simunek, et al. 1998)—within broader integrating frameworks (Manguerra and Engel 1998; SAVABI 1993; Revuelta-Acosta et al. 2021).

Nearly every modelling study attributes residual error to unresolved heterogeneity in hydraulic properties, especially conductivity. Authors seek detailed representation of macropores, fractures, and sand lenses with specific storage and residual saturation (De Schepper, Therrien, Refsgaard, X. He, et al. 2017), depth-varying and dual-porosity parameterisations in layered and paddy soils (Wesström, Ekbohm, et al. 2003; Karandish, Darzi-Naftchali, and Simunek 2018), and denser, depth-resolved sampling—at least one conductivity measurement per hectare with post-construction verification (Palacios-Vélez et al. 2004), more transect points (Messing and Wesström 2006), and horizontal as well as vertical conductivities with hydrophobicity and hysteresis (Tähtikarhu et al. 2025). Anisotropy and long-term property change form a related sub-frontier (Akis 2015; Moustafa and Yomota 1998; Koivusalo, Karvonen, and Paasonen-Kivekäs 1999), and accurate saturated conductivity and drainage-volume/upward-flux relationships, supported by SSURGO-linked subsoil porosity, are emphasised as model inputs (DeBoer and Beck 2006; Schilling, Wolter, et al. 2015; De Schepper, Therrien, Refsgaard, and A. Hansen 2015). Closing the water balance—particularly seepage and groundwater exchange—remains persistent: lateral seepage should vary during simulation (G. Feng, Z. Zhang, P. Lu, et al. 2018), groundwater contributions could enter through double- or residual-storage models (Danesh-Yazdi et al. 2016), and groundwater-table data in objective functions would sharpen inverse recovery of soil properties (Sedaghatdoost, Ebrahimian, and Liaghat 2018; Sedaghatdoost, Ebrahimian, and Liaghat 2019). Improved calibration strategy is widely urged—parameter-estimation software and multiple conceptual models (Boico et al. 2022), pilot-point conductivity fields with proxy machine-learning predictions (R. Schneider, Stisen, and Hojberg 2022), Kling–Gupta objective functions (Brighenti et al. 2023), and transient tests of steady-state assumptions (Schilling, Streeter, Pierce, et al. 2022; Gates et al. 2002; Burkhalter and

Gates 2006). Contemporary needs also span unsaturated modules for disconnected groundwater (Kompanizare et al. 2024), independent layer-to-layer drainage methods under preferential flow (Ojeda et al. 2018), and dynamic frozen-soil models with measured structure (Ulén, Stenberg, and Wesström 2016).

4.5 Nutrient routines, cold-season, and groundwater coupling

Phosphorus and its transport through macropores emerge as the least mature component across water-quality models. Radcliffe et al. (2015) provide the most comprehensive agenda: nested paired field experiments and structural additions across codes—SWAT and ADAPT routing macropore flow directly to drains, a macropore component for APEX, DRAINMOD simulating P losses and sediment-attached P, and HYDRUS parameterising simpler models—with every study quantifying predicted-P accuracy and uncertainty. Consistent with this, T. Guo, Confesor, et al. (2020) recommend physically based tile subroutines, macropore nutrient pathways, and particulate-P losses, Ford et al. (2017) seek understanding of how macropore flow and P respond to tillage, cover crops, and drainage water management, and Bin Shokrana, Ghane, and Z. Qi (2023) identify improved fertiliser-P partitioning and P-only performance statistics, with Skaggs, Youssef, and Chescheir (2012b) anticipating a DRAINMOD phosphorus module. Within SWAT, APEX, and RZWQM2, Williams, K. King, Ford, et al. (2016) argue that incorporating identified flow pathways is critical, H. Qi et al. (2018) recommend a soil-matrix-potential Richards equation and revised particulate-P algorithm, and Moriasi, Rossi, et al. (2012) call for comparing dynamic and constant surface-depression storage. Nitrogen-cycle needs include surface-residue and mobile/immobile-water phases for nitrate transport (Malone, Kersebaum, et al. 2017), complete N budgets in calibration (Gillette et al. 2018), and temporal decay coefficients (Rudra, Negi, and N. Gupta 2005).

Soil N-process representation is repeatedly the weakest link. Priorities include improving nitrification, denitrification, and mineralisation within a better hydrology framework (W. Smith et al. 2020; Fang et al. 2012; X. Hu et al. 2007), and recurring RZWQM agendas—subsurface lateral flow below tiles, simulated soybean N fixation, improved plant N uptake, and reduced winter nitrification (L. Ma, Malone, Heilman, Karlen, et al. 2007; L. Ma, Malone, Jaynes, et al. 2008; L. Ma, Ahuja, et al. 2001; Bakhsh, Hatfield, et al. 2004)—together with revised winter processes in the PNM model (Marjerison et al. 2016) and a more physically based tile-nitrate model using "soft data" and a passive humus pool (J. Qi et al. 2025). Cold-season representation is singled out as inadequate: better freeze/thaw and spring-thaw algorithms (Feyereisen, B. Wilson, et al. 2006), DRAINMOD-N improvement for freezing seasons (C. Yang, Prasher, et al. 2007; Saadat, Frankenberger, et al. 2020; Q. Jiang et al. 2020), a frozen-soil impedance factor and calibratable snow–rain threshold for RZ-SHAW (Z. Li, Z. Qi, et al. 2024), and continued RZWQM winter development (Ahmed et al. 2007; Motarjemi et al. 2021). Coupling drainage models to groundwater is a distinct need: SWAT-MODFLOW-RT3D for reactive transport once salinity is incorporated (Addab and Bailey 2022), water-table-fluctuation and groundwater–surface-water exchange in SWAT to represent riparian nitrate retention (Hoang, Griensven, and Mynett 2017), joint spatial variation in leaching and reduction across systems (Hojberg et al. 2017), and defensible water balances splitting artificial drainage from groundwater flow (Barkle et al. 2021; Elsayed et al. 2023; J. Yang et al. 2008). In arid settings, coupling of water and salt dynamics with unsaturated models, root-growth modules, and large-dataset validation is pressed (Ebrahimian and Noory 2015; Sarangi et al. 2006; X. Han et al. 2025; Y. Liu, Zeng, Ao, et al. 2022; H. Guo et al. 2025).

4.6 Calibration, sensitivity analysis, and uncertainty

How models are calibrated and their parameters interrogated is inseparable from validity. Multi-objective frameworks integrating statistical and graphical diagnostics tailored to the hydrological system are advocated (Bruhn et al. 2025), reliable soil-hydraulic data for the drain layer is emphasised with interacting parameters calibrated and negligible ones fixed (X. Wang, Frankenberger, and Kladivko 2006), and independent calibration against observed discharge is recommended so interception and storage are not conflated with conductivity and drainable porosity (Henine, Jeantet, et al. 2022). Pedotransfer estimates such as Rosetta conductivity should be used cautiously, with further calibration so both water-table depth and discharge are simulated (Malota, Mchenga, and Chunga 2022; Timalsina et al. 2023). A common thread is identifiability—separating parameters that genuinely act from those merely correlated. Automated and global approaches are widely urged: automatic calibration with ensemble Kalman filtering for parameter identification (Warsta et al. 2014; Sadhukhan, Z. Qi, T. Zhang, Tan, and L. Ma 2019; A. Singh 2019b), global sensitivity analysis to capture interactions that local one-at-a-time methods miss (Timalsina et al. 2023; Chelil, Henine, et al. 2022; Kacimov 2000), established bounds for the SWAT sensitivity adjustment factor (Moriasi, Gowda, et al. 2013), and spatially distributed streambed conductance with exchange fluxes as calibration targets and drainage proxies (Karan et al. 2021). Explicit uncertainty treatment is a distinct, underdeveloped frontier: propagation of parameter and measurement error (Alzraiee, Garcia, and Gates 2013), the role of input-data and weather-data quality (Upadhyay et al. 2019; Tao, N. Li, et al. 2021), explicit estimation of P-model prediction accuracy (Radcliffe et al. 2015; Beauchemin et al. 2003), structural uncertainty via multiple water-quality models (Kujawa et al. 2020), probabilistic drainage-water-management effectiveness (Shedekar, K. King, et al. 2021), independent evaluation against withheld data (Schoups et al. 2005), and better statistical treatment of unreplicated paired experiments (Pagliari, Ranaivoson, and Strock 2021; Andrus et al. 2014). The recurring message is that models are too often reported as point predictions when their inputs warrant propagated bounds.

4.7 Model structure, coupling, and machine learning

Beyond recalibration, structural refinements and integrative strategies are judged necessary. Coupling complementary models is a common, largely untested proposal: linking biogeochemical to spatial hydrological models such as SWAT or MIKE SHE for regional prediction (C. Li et al. 2006), coupling nitrate-transport and drainage models with variational assimilation of spring-fertilisation information (Chelil, Henine, et al. 2022), and integrated use of more than one model as more practical than a single code (A. Singh 2019b; Apostel et al. 2021). Machine-learning models raise their own validation requirements the literature is only beginning to confront: hyperparameter tuning and diverse training data for drain detection (Song, Woo, and Q. Yan 2021), neural networks needing training that includes extreme situations and periodic retraining (C. Yang, Tan, and Prasher 2000), and machine-learning surrogates trained across diverse climatic and hydrogeological conditions with transient processes and real-time control (Lei et al. 2026; R. Schneider, Stisen, and Hojberg 2022). Process-model deficiencies are named directly—detailed site characterisation, added non-linear regression dependences, and improved submodels for mineralisation, crop N uptake, and growth under N limitation (Ahuja and Hatfield 2007; Ikenberry et al. 2017), with added complexity the price of predicting snowmelt-thaw drainage timing (Feyereisen, Sands, Strock, et al. 2007). Physical and analytical refinements form a rich set of concrete proposals: numerical models incorporating anisotropy and varying bottom slopes verified against analytical solutions (Lin and Hsieh 2025; Kacimov 2000), reanalysis with the Laplace rather than Boussinesq equation (Bouarfa

and Zimmer 2000), recession equations extended to spatially varying properties (Cooke, Badiger, and García 2001), three-dimensional treatment at heterogeneous sites (Mohanty, Bowman, Hendrickx, and vanGenuchten 1997), distinct mixing-layer depths and partition coefficients for solute transport (Shalit and Steenhuis 1996), reconsidered deep-percolation and shallow-water-table assumptions (Kolars et al. 2019), and strengthened index-system construction with additional soil indicators (Hou et al. 2018).

4.8 Instrumentation and measurement-method validation

Because model validation is only as sound as the data feeding it, uncalibrated flow structures and drifting sensors can masquerade as model inadequacy. In-house calibration of flow-measurement structures is recommended repeatedly: weir calibration for individual applications (L. Christianson, R. Christianson, Lipka, et al. 2019; Napierala 2023), in-house calibrations representing field flow conditions for control structures (Katuwal et al. 2024), stage–discharge equations developed before monitoring for cylindrical structures (Bin Shokrana and Ghane 2021), and post-hoc sensor calibration against grab samples during tracer tests (Oliveira, Maxwell, et al. 2023). Emerging technologies carry explicit validation requirements: a computer-vision flow method needs testing under variable real-world conditions with the Manning–Strickler equation to compensate for uneven velocities (S. Young, M. Han, and Peschel 2023), hybrid RTK–rotating-laser systems address less-than-ideal survey conditions (Freeland, Allred, and Sorochan 2014), and sensor design and scintillometer humidity-stability effects require refinement (Enamorado et al. 2007; X. Zhang et al. 2010). Hydraulic-loss characterisation in drainage conduits remains conspicuously incomplete, limiting design-equation realism: head loss in corrugated-pipe junctions and manholes under free-surface flow (Bayat, Kouchakzadeh, and Azimi 2011), friction loss and transitional cavity-flow conditions (Nédélec and Gay 2008), declining-bend slope effects on air discharge (Gurklys, Rimkus, and Saulys 2008), appropriate Manning coefficients for design (Kouchakzadeh et al. 2011), and strain-gauge characterisation of non-linear HDPE-pipe behaviour (Gaj and Madramootoo 2021). The consistent message across the modelling frontier is that the next decade of progress depends less on new equations than on sustained, well-instrumented, and openly reported field campaigns needed to test the equations already in hand.

5 From Field to Watershed: Mapping Networks and Scaling Response

5.1 Overview

Two linked spatial problems anchor this frontier. First, the physical drainage network—where tiles lie, at what depth and spacing, and how mains connect fields to ditches—is largely unmapped, so uncertainty about the network itself propagates into every downstream hydrologic and water-quality model. Second, even where field processes are well understood, the field is not the scale at which water-quality goals are set; translating instrumented-plot knowledge to the watersheds where nutrient and sediment targets apply remains a genuinely open methodological problem. The two are related: because networks are unknown and because routing, connectivity, and in-stream processes are weakly represented, effectiveness estimates for practices such as drainage water management and phosphorus mitigation cannot yet be aggregated with confidence from field to basin. The priorities that recur across the literature are therefore concrete—non-invasive detection methods with usable practitioner guidelines, shared drainage databases with standardised reporting, intensively monitored small catchments as upscaling bridges, and mechanistic treatment

of the channel-network, connectivity, and depressional processes that mediate the field-to-watershed link.

5.2 Locating, mapping, and cataloguing drainage networks

Because the drainage network is often unknown, detection and mapping form a fast-moving frontier in which methods work under some conditions but lack systematic guidelines. Actual tile locations and measured discharge would sharpen model calibration (Amado et al. 2018), and predictive mapping from imagery needs validation against GPS surveys plus methods that avoid misidentifying noise and that reconnect broken segments (Naz, Ale, and L. Bowling 2009; Schilling, Gassman, et al. 2019). The most explicit data agenda is institutional: report tile diameter, depth, spacing, and mains, and build a shared drainage database jointly maintained by agencies, researchers, landowners, and contractors, with repositories updated as records accrue (S. Zhou and Margenot 2025; Valayamkunnath et al. 2020).

5.2.1 Remote sensing and machine learning

Remote-sensing work seeks urgent development of non-invasive methods for aging systems, including UAV flights combined with ground-penetrating radar (Wienken and Grenzdörffer 2024), and multispectral, thermal-infrared, and machine-learning agendas that define the site conditions under which drain lines and outlets can be mapped and that yield decision-tree guidelines for practitioners (Allred, Martinez, Fessehazion, Rouse, Williamson, et al. 2020; Allred, Martinez, Fessehazion, Rouse, Koganti, et al. 2021; Allred, Martinez, Khanal, et al. 2022). Carlsen et al. (2024) recommend reconsidering the often-cited 2.54 cm minimum rainfall for imagery acquisition and investigating deep learning, multitemporal acquisition, and synthetic aperture radar while expanding public drainage-map datasets—echoed by calls for transformer-based architectures and higher-resolution imagery integrated into hydrologic frameworks (L. Wan et al. 2024) and for reduced tracing uncertainty through multiple interpreters, ground truth, and richer covariates (Redoloza et al. 2023; Harris et al. 2018; Yimer et al. 2023). Optical and radar covariate studies further seek more frequent imaging, repeated SAR with ground measurements, and UAV cover-crop mapping (Tlapáková, Zaloudík, and Kolejka 2017; F. Meunier, J. Meunier, and Cavayas 2001; M. Xu, Lacey, and Armstrong 2018).

5.2.2 Geophysics

Geophysical evaluations remain condition-dependent and call for data from more sites and soils (Allred, Redman, et al. 2005), systematic testing of GPR antenna configuration and orientation (Allred 2013), and prediction of GPR suitability via electromagnetic simulation driven by EMI. Concurrent EMI/GPR surveys and drone or magnetic surveys are recommended, with performance shown to depend on soil, residue, tillage, and wetness (Koganti, Van de Vijver, et al. 2020; Koganti, Ghane, et al. 2021; Jouen et al. 2018; Kratt et al. 2020; Allred, Wishart, et al. 2018). The open question is less whether any single method can detect drains than how to specify, in advance, which method suits a given soil, cover, and moisture state.

5.3 Scaling from field to watershed

The defining open problem is the disconnect between field-scale process knowledge and watershed-scale response. Quantifying the relationships between field processes and watershed response is repeatedly identified as critical for water-quality strategy, motivating explicit field-to-watershed

linkage and upscaling frameworks (Williams, K. King, and Fausey 2015; Williams, K. King, and Fausey 2017; Z. Wang, T. Zhang, Tan, and Z. Qi 2020). Small, intensively monitored catchments are proposed as the empirical bridge for upscaling and model validation (Kyllmar et al. 2023; Basu et al. 2010), and effectiveness estimates for drainage water management and phosphorus mitigation should be scaled up from field to watershed (Z. Wang, T. Zhang, Tan, Xue, et al. 2021; Ross et al. 2016; Shedekar, K. King, et al. 2021).

The hydrologic effects of tiling remain inadequately characterised at watershed scale and must be pursued through combined modelling and field data: attending to seasonal streamflow recessions in tiled versus non-tiled watersheds (Schilling and Helmers 2008; Schilling, Jindal, et al. 2012), to events of all sizes across landscapes (Sloan, Mantilla, et al. 2017), to routing between spatially variable depressional stores and flow accumulation across river networks (Sloan, Basu, and Mantilla 2016; Fonley et al. 2021), and to the decoupling of row-crop cover from artificial drainage and of surface from subsurface contributions (Boland-Brien, Basu, and Schilling 2014). Streamflow prediction in tiled landscapes is itself difficult, raising the unresolved question of whether curve-number or more process-based methods are required (Wessel et al. 2022; Thomas et al. 2016; Rahman et al. 2014). Whether a representative catchment scale transfers elsewhere must be established against additional watersheds (Basu et al. 2010; Thomas et al. 2016), and newly proposed physically based tools such as the Advanced Terrestrial Simulator and HYDRUS-2D warrant scaling and transferability tests across soils, topography, land-use change, and climate (Rathore, Sloan, and Painter 2025; Naderi et al. 2025; Basu et al. 2010).

5.4 Channel-network, connectivity, and drain-to-stream processes

Interactions between drains and the receiving channel network are a frontier in their own right. Open questions include the fate and transport of tile effluent after it reaches surface drains (Clement and Steinman 2017), drain–river and drain–ditch interactions (Grangeon et al. 2021), and field-scale monitoring of how tile flow affects downstream discharge, stream power, and channel geomorphology (S. Zhou and Margenot 2025; X. Zhou, Helmers, and Z. Qi 2013).

Channel-network, in-stream, and sediment processes are underdeveloped in watershed models. Networks should be modelled as interdependent reaches with explicit flow paths and travel times (Giraud et al. 1997; Williamson et al. 2025); in-stream biogeochemistry and hillslope–riparian interaction need stronger representation (Schilling, Jones, Seeman, et al. 2012; Basu et al. 2010); and gully, streambank, and non-field sediment sources require mechanistic treatment within catchment-scale sediment budgets (Maalim and Melesse 2013; Dalzell and Mulla 2018; Walling et al. 2002).

Connectivity metrics are increasingly flagged as management-relevant. Mapping surface ditches may be more efficient than locating in-field drains, and future phosphorus-risk models should incorporate ditch connectivity and lag times (Moloney, Fenton, and Daly 2020; Opoku et al. 2024), supported by improved connectivity algorithms and monitoring-based thresholds for lowland catchments (Calsamiglia et al. 2018). Finally, drained closed depressions and potholes are singled out as poorly understood: they demand process study at additional locations, quantification of runoff–tile–antecedent-moisture relationships, and data on blind-inlet longevity (Martin, Kaleita, and Soupir 2019; Williams, Livingston, et al. 2020; Feyereisen, Francesconi, et al. 2015). Across these threads, the common gap is a lack of coupled, connectivity-aware representations linking individual drains to the reaches and depressions that ultimately govern watershed water quality.

6 Managing Nutrient Losses: Nitrogen, Phosphorus, and Cropping Systems

In-field nutrient management remains the primary lever for reducing nitrate and phosphorus (P) losses from tile-drained land, yet the levers themselves are better described than optimised. The frontier is less about which practices help than about the metrics, mechanisms, crediting rules, and temporal targets that would let managers set the right rate, form, and timing for a given soil, climate, and cropping system. Z. Ma et al. (2025) frame the programme broadly, arguing that far more work is needed to understand the tradeoffs among productivity, nutrient-loss reduction, and ecosystem services in tile-drained systems. This section synthesises the open questions for nitrogen, for phosphorus, and for cropping-system change; model-algorithm gaps, greenhouse-gas tradeoffs, edge-of-field treatment, and spatial bundling are treated only in passing here and developed elsewhere in the review.

6.1 Refining nitrogen management and the timing of loss

Refining fertiliser nitrogen (N) is the most consistently identified priority. ROSSI, CIAVATTA, and ANTISARI (1991) call for careful evaluation of fertiliser timing, rate, form, and placement, and S. Zhao et al. (2000) for strategies that raise N-use efficiency in corn through optimal rate, timing, and form. Randall and Mulla (2001) add crediting of previous legumes and manure and improved manure management, while Dattamudi et al. (2020) conclude that application schemes need reevaluation toward lower corn rates, precision agriculture, and split, properly timed applications—echoing Elmi, Madramootoo, Egeh, Dodds, et al. (2002), who see reductions from matching rate and timing to crop uptake. Optimising N rates is expected to retain primary research and regulatory focus, with accurate 4R implementation requiring site-specific knowledge to balance agronomic and environmental goals (L. Christianson and Harmel 2015).

6.1.1 Next-generation rate tools and decision support

Bandura et al. (2025) propose concrete standards for the next generation of N-rate tools: developed against the economically optimum N rate (EONR), adequate if they predict need within 10 kg N ha^{-1} of EONR, and environmentally unacceptable if they exceed it by more than 30 kg N ha^{-1} at an N:corn price ratio of 5.6. W. Liu, Yuan, and Koropeckyj-Cox (2021) call for recommendations that weigh environmental and economic benefits, case-by-case cost analyses, and coordinated agency action to promote 4R adoption, and Banger et al. (2018) for experiments across wider soils and management to identify yield tradeoffs from rate reductions—shifting emphasis toward cover crops and edge-of-field practices where yield-loss risk appears. Comprehensive tools integrating weather forecasts, management history, and local field parameters, with regional agronomic-optimum-N-rate benchmarks (Maas et al. 2022), precision application by soil-N availability (R. Gupta, Bhattarai, Dokoohaki, et al. 2023), better crediting of non-fertiliser N and variable-rate models incorporating remote sensing (Dinnes et al. 2002), and low-cost proxy variables for soil N-supply capacity (Cassman et al. 2025) are all advanced. Local decision approaches also need refinement: late-spring nitrate-test programmes are framed as a start to be replaced as research emerges (Jaynes, Dinnes, et al. 2004), with grain-protein monitoring and risk insurance for adopters (Karlen et al. 2005) and weather-based tools to reduce soil-testing needs (Malone, Jaynes, L. Ma, et al. 2010). Jaynes (2013) seek to pinpoint where between the V12 and V16 stages sidedress N ceases to benefit, and Jaynes (2015) note sidedressing can be paired with soil tests or crop N-deficiency sensing for real-time adjustment; comprehensive volatilisation studies and data from data-poor regions are further gaps

(Chatterjee 2020).

6.1.2 Timing, form, and application method

Method-level questions remain open. Malone, L. Ma, et al. (2007) highlight quantifying split versus single spring applications, and Hernandez-Ramirez et al. (2011) note it is unknown whether reduced-rate fall manure could match sidedress environmentally, pointing to nitrification inhibitors and more complex rotations; Elmi, Madani, et al. (2005) stress quantifying all major N loss pathways simultaneously. Product-level frontiers include enhanced-efficiency fertiliser with a nitrification inhibitor tested alongside drainage water management (Kaur and Nelson 2024), NH_4^+-N stabilisers and cover-crop effects on NH_4^+-N dynamics (Vincent et al. 2025), and fertigation of silage corn with its crop-injury and pathogen tradeoffs (J. Gamble et al. 2018). Evening out manure application in high-livestock catchments (Bechmann 2014) and comparing N-use consequences under global-warming projections (Haghnazari et al. 2020) round out this cluster.

6.1.3 Crediting residual, mineralisable, and legacy nitrogen

A recurring, more specific gap is crediting N supplied by soil organic matter, prior legumes, manure, and land-use legacies. Bakhsh, Kanwar, Pederson, et al. (2007) and Bakhsh, Kanwar, and Karlen (2005) call for assessing proper credit from mineralisation of the organic-matter pool and soybean residue before applications, and Randall and Mulla (2001) for improved soil-N tests to estimate mineralisable and carryover N—echoed by a spring test after dry years (RANDALL and IRAGAVARAPU 1995) and pre-sowing mineral-N measurement (Conrad and Fohrer 2009). Systems transitioning from perennial cover need second-year credits after grass conversion (Huggins, Randall, and Russelle 2001) and care after alfalfa (ROBBINS and CARTER 1980), while irrigated vulnerable soils call for reduced early-year N with residual-nitrate crediting (Derby, Casey, and Kinghton 2009). Legacy N is itself underexplored: a clearer definition is needed to apportion losses (Williams, Penn, and McAfee 2022), and the coupled biogeochemical and hydrological processes regulating legacy-N formation and its effect on nitrate export require dedicated study (Y. Hu et al. 2024). Reconsidering fall application under warmer winters (Gentry, David, and McIsaac 2014), resin-membrane strips to characterise soil-N cycling (Masri, Asher, and Piwarski 2024), and better long-term flow-weighted-N predictions for benchmark soils (Skaggs, Youssef, and Chescheir 2012a) complete the agenda.

6.1.4 The temporal window of loss

A distinctive frontier is the temporal concentration of losses and the need for management targeted to that window. Bakhsh, Kanwar, Pederson, et al. (2007) argue strategies must reduce early-spring nitrate leaching, and Bakhsh, Kanwar, and J. Baker (2010) extend this to March–June while flagging the temporal distribution of growing-season rainfall. Assessment of nitrate status and tile loss during April, May, and June warrants study (Nance and Gibson 2007), and a winter-hardy cover crop ahead of corn with a shift of fall N to spring may best cut loss in intensively tiled watersheds (Gentry, Green, et al. 2024). For outlet management, Skaggs, Breve, and Gilliam (1995) recommend selectively managing outlets to reduce peak nitrate losses, especially after drought-reduced yields, and Elmi, Madramootoo, Egeh, and Hamel (2004) and Elmi, Madramootoo, Egeh, Dodds, et al. (2002) note that keeping controlled drainage operational in early spring and late fall—when losses are severe and drainage is not needed for production—could achieve further gains.

6.2 Phosphorus metrics, mechanisms, and transport

Phosphorus in subsurface drainage is repeatedly flagged as under-researched relative to N. Hertzberger et al. (2019) call for greater effort on P transport, and Yuan et al. (2022) for research on conservation-practice effectiveness in reducing dissolved-P loss and the causes of its annual variability. Pease, K. King, et al. (2018) identify pedologic controls on P transport as key needs, and Oliveira, Ventura, et al. (2022) recommend refining loss frameworks through within-field studies and a better understanding of the "where" and "when" of drainage P loss. A first-order requirement is to resolve partitioning and sources: K. King, Williams, et al. (2015) call for enhanced P routines and simultaneous measurement of P losses in surface runoff and subsurface drainage to determine partitioning, Williams, K. King, Ford, et al. (2016) for differentiating soil versus fertiliser sources of tile-drained P, and H. Qi et al. (2018) for measuring the size distribution of particulate-P carriers.

6.2.1 Beyond agronomic soil-test P: saturation and sorption metrics

A consistent theme is that agronomic soil-test P (STP) is insufficient for environmental risk. Sharpley (1995) argue that environmental tests estimating bioavailable P and sorption capacity should replace agronomic tests on high-risk soils. Hooda et al. (1999) propose that degree of P saturation (DPS) may better estimate loss potential than extractable P such as Olsen-P, and L. Nguyen and Sukias (2002) call for critical DPS values calibrated across sediment-P tests and equilibrium P concentrations. Osterholz, Hanrahan, and K. King (2020) recommend assessing fields in the 100–400 mg kg^{-1} STP range and examining sorption saturation, macropore formation, and mineralogy, while P-adsorption capacity should precede developing organic soils for cropping (MILLER 1979). McDowell, Sinaj, et al. (2001) propose that P isotope-exchange kinetics could explain and eventually predict loss across a wide range of STP.

6.2.2 Desorption, colloidal, and particulate transport

The sorption/desorption interface and particulate pathways are poorly resolved. Mumbi et al. (2025) regard quantifying when and how varying water sources and fluxes desorb and adsorb P as critical to predicting subsurface DRP transport, and Flores, Bailey, and Harmel (2021) note the need to understand particulate settling and P desorption. Uusitalo, Turtola, et al. (2003) argue that mitigating bioavailable particulate P requires reducing particle mobilisation across the whole field surface and that controlling diffuse DRP is a longer-term task than reducing erosion, with the availability of eroded P to algae still limited (Ulén, Bechmann, et al. 2007). R. Wilson et al. (2019) highlight quantifying the colloidal-P contribution to DRP, and Norgaard et al. (2013) recommend including both dissolved and colloid-associated leaching parameters in field-scale risk assessment. Sediment-P release under alternating waterlogged–drying regimes (L. Nguyen and Sukias 2002), P fractionation under polyacrylamide (T. Jiang et al. 2010), and combining columns, plots, and catchments to link particulate loss to macroporosity (Laubel et al. 1999) are further priorities.

6.2.3 Management effects, events, and legacy phosphorus

Uusitalo, Lemola, and Turtola (2018) note that assessing the pace of P stratification needs more data and leave open whether no-till adoption as a eutrophication measure is justified. Ni, Yuan, and W. Liu (2020) call for studies of optimal STP under changing climate to address STP–precipitation interactions, and Y. Wang, T. Zhang, et al. (2018) ask whether pre- or post-application tillage decreases tile DRP and whether findings extend to manure injection, with immediate post-application

manure risk still poorly understood (Madison et al. 2014). Brendel et al. (2019) recommend designing practices that reduce P losses during frequent small events, and Schilling, S. Kim, et al. (2017) refining organic-P/total-P ratios and quantifying wastewater-treatment contributions. Jarvie et al. (2017) ask how much of a 40% reduction target 4R practices alone could achieve and warn that drawing down legacy P may be increasingly costly and intrusive; cost-effective hierarchies prioritising lower Olsen P and targeted alum (McDowell 2015) and P removal by grass harvesting (T. Zhang, Tan, Z. Zheng, and Drury 2015) are advanced. Fragmented reporting also hampers synthesis: L. Christianson, Harmel, et al. (2016) call for field studies documenting multiple P species and further developing the Drain Load database, Macrae et al. (2019) for understanding hydroclimatic event drivers and tillage or placement effects, Tanga et al. (2025) for monitoring particulate and dissolved forms with site-specific stoplog depths, T. Zhang, Tan, Y. Wang, et al. (2017) for new soil-P testing under no-till, and Spiese, M. Bowling, and Moeller (2023) for novel practices addressing nitrate and DRP simultaneously.

6.2.4 Phosphorus in organic soils

Phosphorus in cultivated organic soils is singled out as poorly understood. Grenon, De Sena, et al. (2022) call for mechanistic studies of the pathways, hot spots, and hot moments of P transfer from agricultural organic soils, and Grenon, B. Singh, et al. (2021) propose a concrete agenda using isotope tracing, XANES, and ^{31}P NMR alongside sequential fractionation, microbial-turnover assessment, and long-term drainage-water-management studies. Nutrient leaching from abandoned peat fields, and the true nature of H_2SO_4-P, remain poorly studied (Höyhtyä et al. 2025).

6.3 Cover crops, perennials, and rotational diversification

6.3.1 Cover crops: agronomic, environmental, and modelling frontiers

Cover crops—especially winter rye—attract the largest concentration of open questions. Kladivko, Kaspar, et al. (2014) call for research and technical assistance to overcome adoption obstacles and for field experiments across more Midwestern soils, and Birru et al. (2023) and X. Zhao et al. (2016) stress that existing evidence is drawn from too few, geographically clustered sites and must be extended across more species, soils, and climates. Agronomic priorities include tools to promote rye in colder regions (Waring, Lagzdins, et al. 2020), establishment strategies under temperature and precipitation extremes (Bawa et al. 2023), evaluation of more cultivars to reduce yield variability (Kaspar, Jaynes, Parkin, and Moorman 2007), breeding for cold hardiness (Dinnes et al. 2002; J. Liu, Macrae, et al. 2019), whether yield drag reflects poor planter performance through residue (B. Dougherty et al. 2020), and better management of kura clover as living mulch (Z. Qi, Helmers, R. Christianson, et al. 2011; Z. Qi, Helmers, and Kaleita 2011). On environmental mechanisms, authors seek research on the spring release of scavenged N and its synchronisation with crop growth (Lacey and Armstrong 2015), adjustable seeding rates (Roth et al. 2018), combined cover-crop and manure effects on subsurface DRP (Askar, Hanrahan, et al. 2023), and optimisation of cover-crop services against disservices to the water balance (Meyer et al. 2019). J. Liu, Macrae, et al. (2019) call for coordinated common experiments to quantify winter-cover P contributions, longer-term studies remain scarce (Kaspar, Jaynes, Parkin, Moorman, and Singer 2012; Martinez-Feria et al. 2016), growing cover crops longer in spring may prevent N_2O increases (Preza-Fontes, L. Christianson, and Pittelkow 2023), and because over-winter effects differ for N versus P, practices should target individual pathways (McNeill et al. 2026)—with Krueger et al. (2013) recommending practices beyond rye where subsurface nitrate stayed high and rye had no effect on DRP. Cover-crop modelling is a distinct frontier: validated rye growth and plant-N- uptake algorithms (Feyereisen,

Sands, B. Wilson, et al. 2006; Malone, Huth, et al. 2007), dedicated winter-rye modules and calibration (Malone, Jaynes, Kaspar, et al. 2014; Chatterjee et al. 2025), and correction of cover-crop and frozen-soil weaknesses (Dietzel et al. 2016; R. Gupta, Bhattarai, Coppess, et al. 2022; R. Gupta, Bhattarai, Kalita, et al. 2023) are sought, with detailed algorithm gaps developed in the modelling section.

6.3.2 Perennials, bioenergy, and rotational diversification

Perennial and land-use-change options feature prominently. Poorly drained depressions are proposed for perennial bioenergy crops such as *Miscanthus* × *giganteus* or wetland restoration (W. Huang et al. 2023), and keeping soil covered year-round with perennial ley is judged the most effective way to mitigate N losses, with the P focus on decreasing surface runoff (Norberg et al. 2022). Additional data on alfalfa's effect on subsurface discharge during large events (Arrueta, K. King, et al. 2025), a clear causal link between perennial crops and nutrient loads with policies rewarding environmental services (Arrueta, Hanrahan, et al. 2022), robust water-quality datasets for perennialised systems (G. Wilson et al. 2023), and large-scale bioenergy-scenario assessment with cost–benefit analysis to alleviate Gulf hypoxia (T. Guo, Cibin, et al. 2018; Woodbury et al. 2018) extend the agenda. Rotation and diversification questions recur: accounting for losses on a rotation rather than annual basis (X. Zhao et al. 2016), the paucity of research on desirable rotation crops in two-crop systems (Pasley et al. 2021), partial conversion of cropland to pasture or wetland to meet nitrate targets (Randall and Mulla 2001; Gowda, Dalzell, and Mulla 2007; Nangia et al. 2008), and explicit joint evaluation of controlled drainage, rotation, cover crops, and N management (L. Ma, Malone, Heilman, Jaynes, et al. 2007). Novel options—triticale with red clover (Nance and Gibson 2007), camelina–soybean relay-cropping (Emmett et al. 2022), and fertilised, harvested rye for revenue (Malone, Obrycki, et al. 2018; Malone, Radke, et al. 2023), and diverse systems benchmarked against the two-year rotation (Gentry, C. Mitchell, et al. 2025) —are all advanced. Dinnes et al. (2002) call broadly for perennial cash-cropping systems that maintain returns while reducing leaching, and Jaynes, Colvin, et al. (2001) conclude that dramatic management change on artificially drained soils will be required before corn production can be considered sustainable.

7 Water-Table Management and Drainage-System Design

The frontier in drainage water-quantity management is defined by a single shift: from passive, one-way conveyance toward active, location-specific control of the water table and the coupled irrigation–drainage balance. The enabling ideas—water-table-control structures, automation, forecast-based operation, and re-evaluated physical design—are largely in place, but each remains a hypothesis awaiting the long-term, multi-site field and simulation evidence needed to quantify its yield and water-quality trade-offs. What follows organises these open questions around active water-table control, the control systems that would automate it, and the design criteria that would support it.

7.1 Controlled drainage, subirrigation, and drainage water management

Integrating controlled drainage (CD) with irrigation is repeatedly framed less as settled practice than as a research frontier: the recurring call is to move from passive, "unilateral" drainage toward active control of the water table and the coupled irrigation–drainage balance (Ayars, Christen, and Hornbuckle 2006; Kesicka et al. 2023; Ritzema, Abdel-Dayem, et al. 2023). Ayars, Christen, and Hornbuckle (2006) argue that water-table-control structures should be required in new designs and considered for retrofits, and that new criteria and management methods are needed—claims that

require long-term field studies, possibly combined with modelling, to validate—echoed by calls to build in control options and expand changeable CD networks (Dukhovny et al. 2018; Mahmoud et al. 2021; Strock, Sands, and Helmers 2011). The quantitative payoff of active management remains open: Watkins et al. (2024) hypothesise that active outlet management during extremely wet periods could minimise wet-stress and raise yields; Saadat, L. Bowling, Frankenberger, and Brooks (2017) emphasise the cost–benefit trade-off; and Helmers et al. (2012) suggest that CD yield reductions could be mitigated by greater management while weighing installation cost, management burden, and applicable acreage. Seasonally differentiated control—free drainage in wet seasons, CD in dry seasons, with supplemental irrigation for extremes—is an explicit hypothesis to test (Asgari et al. 2021).

Coordinating irrigation with drainage control is a persistent unsolved problem: irrigation must be reduced when CD is implemented, requiring coordination among authorities and farmers (Wahba, Christen, and Amer 2005; Ritzema and Schultz 2011; Ritzema 2016), and CD requires genuinely integrated management with careful site selection (Dukhovny et al. 2018). Subirrigation embedded in regional water systems is a distinct, largely unresolved frontier: regional water needs should be known before responsible incorporation, with optimised pumping and cautious use of alternative resources (Wit, Ritsema, et al. 2022), and SWAP-type modelling can pre-screen sites and forecast consequences before implementation (Wit, Huijgevoort, et al. 2024). Seepage effects on CD performance need coordinated experimental and modelling research, including guard lines to evaluate lateral seepage (Youssef, Y. Liu, et al. 2021; Pease, Fausey, et al. 2017; Ale et al. 2009), while excess-moisture stress, spring drawdown, and spring nitrogen loading remain priorities (Craft et al. 2018; Lahdou et al. 2019). Finally, longer-term studies must disentangle whether soil-property changes under drainage–subirrigation arise from the practice itself or from irrigation-water quality (Kaur, Nelson, G. Singh, et al. 2023), and complete water and nutrient budgets within and across field boundaries—plus effects on runoff, preferential flow, and recharge—remain to be tracked (Shedekar, K. King, et al. 2021; Ross et al. 2016; K. King, Hanrahan, et al. 2022).

7.2 Automation, forecast-based, and data-driven control

Automation and data-driven control are identified as the next technological step, but the control systems themselves are still to be built and validated. Youssef, Strock, et al. (2023) call for a new generation of control structures and decision-support tools, sensor-based systems tailored to shallow-water-table soils, and systems that adapt to changing weather; Ayars and Evans (2015) extend this to adaptive-control decision support. Concrete methodological agendas include dynamic modelling and data assimilation of the field water balance, feasible formulations of the control problem, and fusion of control automation with physical infrastructure (Ikonen, Liedes, and Vasara 2023; Isomäki et al. 2024), automatic water-depth control for aerobic rice (Arif et al. 2021), and—anticipated decades earlier—farm decision aids that jointly incorporate water quantity, quality, crop growth, and economics (THOMAS et al. 1995).

Anticipatory, forecast-based operation is the emerging synthesis of these threads. Forecast-driven management should incorporate evapotranspiration, subirrigation, and spatially variable seepage, and must explicitly address forecast uncertainty (Van de Craats, Van der Zee, and Leijnse 2021); collector-ditch water levels and control schedules should be optimised (Salla, H. Salo, and Koivusalo 2022; Strock and Ranaivoson 2025); and control should be paired with local crop and climate characteristics without overlooking the risk that rapid water-table decrease raises irrigation demand in dry summers (Güngör and B. Bahçeci 2023; Dou, Shi, R. Li, Miao, J. Yan, and F. Tian 2023; Golmohammadi et al. 2016; J. Kim et al. 2023; Z. Wang, Shao, et al. 2020). The open questions are thus as much about coordinating forecasts, sensors, and structures into a single

adaptive system as about any individual component.

7.3 Novel structures, design criteria, and drainage intensity

Novel structures and materials define a first set of open questions: improved HYDRUS-2D boundary conditions and field verification of siphon drainage (Ren, S. Wang, P. Yang, and Tao 2023), long-term measurement of working time against construction cost (Tao, J. Liu, Su, et al. 2025), and new structures that adjust water-holding capacity while inhibiting rapid vertical and bypass flow (Y. Li et al. 2021). Multi-level and bi-level configurations warrant study of actively managed shallow and deep drains (Hornbuckle, Christen, and Faulkner 2007; A. Verma, S. Gupta, K. Singh, et al. 1998; Nelson and Meinhardt 2011), and envelope, perforation, and screen design remain unresolved across gradients, soils, and textures (Alavi et al. 2021; Z. Wu et al. 2022; Oyarce, Gurovich, and Duarte 2017; Haj-Amor and Bouri 2019). Optimal layout and burial-depth determination in saline and problem soils recur in recent studies, which stress ditching depth, restrictive-layer breaking, amendments, and construction cost within multi-scheme decision models (F. Tian, Shi, et al. 2023; F. Tian, Miao, Shi, R. Li, Dou, Duan, and W. Feng 2024; F. Tian, Miao, Shi, R. Li, Dou, Duan, J. Liu, et al. 2023; S. Zhang et al. 2024; Zhai et al. 2023; D. Han et al. 2023; Y. Qian et al. 2025; Ren, S. Wang, P. Yang, Tao, and H. Chen 2021; S. Zheng et al. 2025; Serrano-Coronel et al. 2018; Zegeye et al. 2021). Analytical advances are also sought—formulas for sloped and multilayer soils (Tao, S. Wang, D. Xu, and Qu 2016; Tao, S. Wang, D. Xu, Guan, et al. 2019), partially penetrating and trapezoidal ditches (Chahar and Vadodaria 2010; Afruzi, Nazemi, and Sadraddini 2014), and mass-transfer and three-dimensional flow effects (J. Zhang, Ye, and C. Lu 2024)—alongside CAD/GIS-based design software and standardised drainage-coefficient definitions to enable comparison (GALLICHAND, MARCOTTE, and PRASHER 1992; Skaggs 2017).

Many site studies conclude that established design criteria themselves require re-evaluation. New criteria that protect crops while minimising drain-water salinity and volume are sought (Christen, Ayars, and Hornbuckle 2001), along with updated depths and spacings at specific sites (Muhammad, Ibrahim, and El-Sayed 2021; Malota and Senzanje 2016; I. Bahçeci and Nacar 2009; Kahlown and Azam 2002), review of recommended spacings paired with controlled-traffic farming (Mante, Ndulue, et al. 2022; Mante, Ranjan, and Bullock 2018), cost-effectiveness of narrow spacings in claypan soils (Nelson 2017), and standard depth-and-spacing guidelines tested in situ before extension (Haj-Amor and Bouri 2019). Design should further account for changing crop water needs, climate, and irrigation modernisation (Rui, Goller, and Kladivko 2024; Akram et al. 2013; Ziaeifar et al. 2025; Ritzema, Abdel-Dayem, et al. 2023; Ritzema, Wolters, et al. 2007; Ritzema, Nijland, and Croon 2006; Ritzema and Schultz 2011).

Drainage intensity emerges as a systems question rather than a per-field specification. Reduced spacing would require increased network capacity and heighten reliance on best management practices (R. Singh et al. 2009; Dialameh, Ghane, and Youssef 2025); a systems approach should integrate drainage, crop–soil processes, and nutrient-loss reduction (Castellano et al. 2019); and DRAINMOD-based analysis should develop optimum rates and spacings across locations (Strock, Sands, and Helmers 2011). Which soil parameters are most critical, and what confidence intervals each requires, remain fundamental unanswered questions (Ayars and Evans 2015; SHIRMOHAMMADI, WENBERG, and RITTER 1995; HARDJOAMIDJOJO and SKAGGS 1982). Underlying all of these is a methodological prerequisite: design guidance must rest on long records, since drainage design should not rely on short-term simulations (X. Wang, Mosley, et al. 2006), and long runs with hourly input are needed to assess wider spacings, interannual variability, and shallow-water-table control (Ale et al. 2009; Häggblom et al. 2019; Malota and Senzanje 2015; H. Salo, Mellin, et al. 2019; Kladivko and L. Bowling 2023; Chang et al. 2012).

8 Edge-of-Field Treatment Technologies

8.1 Overview

Edge-of-field treatment structures—denitrifying woodchip bioreactors and their companions (constructed wetlands, saturated riparian buffers, and phosphorus-removal structures)—have accumulated abundant performance demonstrations yet remain defined more by open design questions than by settled rules. Bioreactors remove nitrate by microbially mediated denitrification, but their performance is bounded by low water temperature, variable hydraulic retention time (HRT), and the biogeochemical evolution of the woodchip medium as it ages. The frontier clusters into interlocking lines: how to size, scale, and standardise design; how to resolve the internal hydraulics and tracer methods on which design rests; whether the microbial engine can be deliberately enhanced; how to select, amend, and manage reactive and phosphorus-sorbing media; how to characterise multi-decadal aging, clogging, and construction effects; how pumped and coupled configurations extend the design space; and how companion practices complement or substitute for bioreactors. A recurring meta-message is the need for standardised design procedures, broader performance-metric reporting, and long-term data across climates, landscapes, and maturation stages so that watershed-scale nutrient-reduction goals can be met (L. Christianson, Cooke, et al. 2021; Maxwell, Cooke, and L. Christianson 2022; Bell et al. 2015). The evidence is drawn overwhelmingly from site-scale field and laboratory studies concentrated in the U.S. Midwest; the by-products of reductive treatment (greenhouse gases, dissolved organic carbon export, methylmercury) are treated here only as design constraints, their downstream accounting deferred to the consequences synthesis.

8.2 Sizing, scaling, and standardised design

A central frontier is reconciling site-level effectiveness with watershed-scale load removal. L. Christianson, Hay, et al. (2026) argue for better balancing bioreactors that are as effective as practical at the site while still removing as much load as possible; they flag the untested performance of *undersized* bioreactors serving larger drainage areas, propose greater design flexibility to scale the practice, and caution that additional empirical data are needed to strengthen the loading-density relationship and that performance data from full-size bioreactors beyond Illinois are required before prediction models are used for design. For hydrologically flashy sites, H. Dougherty et al. (2020) propose sizing designs upward and repeatedly call for simpler, standardised "plug-and-play" design procedures; standardisation is echoed at the policy level by L. Christianson, Cooke, et al. (2021), who recommend refining the Conservation Practice Standard and streamlining the design process. The adequacy of residence-time (T)-centric design is itself questioned: Oliveira, H. Dougherty, et al. (2023) advocate field-scale assessment of baffles in wide bioreactors to raise hydraulic loading while preserving efficiency, and more tracer testing of non-standard media coupled to nitrate monitoring and modelling. Foundational validation remains incomplete—further field-scale validation of the design procedure against beds of differing design (L. Christianson, R. Christianson, Helmers, et al. 2013), validation of hydraulic models at additional denitrification-bed sites with conservative isotopic tracers (Ghane, Fausey, and Brown 2015), region-specific parameterisation of flow, seasonality, and microbial limits for new settings such as Lithuania (Povilaitis and Matikiene 2020), and sorption experiments underpinning tracer-based design (Ghane, Feyereisen, and Rosen 2019a).

8.3 Hydraulics, retention time, and tracer methods

HRT is repeatedly identified as the master variable linking hydraulics to removal, yet its treatment in design guidance is imperfect. Moorman et al. (2015) note that research on the HRT–nitrate-

removal relationship would sharpen sizing criteria, and Povilaitis, Rudzianskaite, et al. (2018) call for proper optimisation of HRT effects through both laboratory- and field-scale study across flow conditions. In a global synthesis, L. Christianson (2025) recommend methodological advances: assessing varying HRT by changing flow rate versus sampling along the reactor length; greater transparency in estimating woodchip porosity; more consistent reporting of dissolved oxygen, pH, and dissolved organic carbon; and machine-learning refinement of HRT–removal-efficiency estimates as data accumulate. The internal hydraulics of woodchip media are poorly resolved: Ghane, Feyereisen, and Rosen (2016) argue that the non-Darcy flow characteristics of woodchips should be exploited to revise NRCS design guidelines. Tracer methodology is a specific bottleneck. Ghane, Feyereisen, and Rosen (2019b) report anomalously low bromide recovery and recommend testing for organobromine in woodchips, microscopic and excavation-based examination of physical bromide entrapment, and either accounting for retardation or adopting non-sorbing conservative tracers such as isotopes; Ghane, Feyereisen, Rosen, and Tschirner (2018) likewise recommend tracer tests to compare the hydraulic performance of retrofitted beds. Reliable tracer methods are thus a prerequisite to benchmarking hydraulic performance at all.

8.4 Enhancing the microbial engine

The defining enhancement frontier is whether denitrifiers can be inoculated into woodchip bioreactors to raise removal where cold temperatures limit biological activity. Jang et al. (2019) propose that bioaugmentation might increase low-temperature removal but note that genome completion is needed to identify the nitric-oxide reductase of a candidate *Cellulomonas* inoculant. E. Anderson et al. (2020) frame the first-order uncertainties: whether inoculated strains enhance field removal, how long they survive in bioreactors, and whether carbon addition boosts both inoculated and indigenous denitrifiers. Survival and persistence recur globally, with H. Wang and Ishii (2025) calling for field-scale methods that minimise inoculum loss and for exploration of emerging targets and consortia—nitrate-dependent anaerobic methane oxidation (n-DAMO), biostimulation to favour inoculated bacteria, and algal–bacterial granular sludge. Terminal denitrification steps in fungal strains, including confirmation of nitrous-oxide reductase and reduction of nitrate to dinitrogen under bioreactor conditions, remain to be verified (Aldossari and Ishii 2021), and microbial amendments require replicated pilot-scale testing before upscaling (Jeglot et al. 2022; C. Ma et al. 2021). Against this enthusiasm, Hathaway et al. (2015) offer a testable counter-hypothesis—that optimisation is more likely from refinements to design and management parameters that shape environmental conditions than from direct microbial manipulation—a useful null hypothesis for field trials.

Rational manipulation presupposes mechanistic understanding of the resident community, which is incomplete. Hartfiel, Schaefer, et al. (2022) argue that bioreactor microbiome study should draw on soil and wetland ecology; that controls on cellulose- and lignin-metabolism genes warrant attention; that dissimilatory nitrate reduction to ammonium and anammox have been observed but understudied; and that varying substrates to select for differing communities is a promising route to more consistent performance and reduced by-product formation—the mechanism by which HRT shapes the community, and thereby by-product production, being a specific gap. Complementary needs include managing bed pH for acidic drainage and conceptualising regional controls on in-bed nitrogen removal (Bock, Coleman, and Easton 2018), routine monitoring of nitrite to avoid its accumulation in effluent (Kouanda and Hua 2021), and characterising the selective pressure of oxygen versus nitrate load on community assembly (Schaefer et al. 2021). The emissions accounting of these by-products is treated in the consequences synthesis; here they define a design and monitoring constraint.

8.5 Reactive media, carbon sources, and structural materials

Beyond conventional woodchips, alternative fill and carbon-management strategies are incompletely characterised. Law, Slade, et al. (2023) encourage field studies of corncobs capturing spatial and temporal variability, the nitrogen-removal-rate range, and whether a lower maximum HRT is needed, while Porter et al. (2015) call for study of organic-carbon availability at different depths and moisture contents, so that substrate choice or internal water levels could be tuned. Because woodchip carbon fuels denitrification, its release kinetics and pre-treatment are active frontiers: the longevity of composting effects and the biodegradability and release rates of woodchip DOC remain to be quantified (Kouanda and Hua 2021), and startup carbon losses raise questions about recycling the initial flush or pre-washing chips before installation (Abusallout and Hua 2017). Synthetic-carbon dosing is unresolved— the source, frequency, and amount needed to optimise performance under dynamic inflow while avoiding excess DOC and biofilm clogging (Rivas et al. 2020), and the hybrid application of synthetic carbon in low-temperature-tolerant reactors (F. Zhang et al. 2021).

A distinct materials frontier concerns drain envelopes and structural fill. Rice-husk envelopes require field testing of combination ratios and long-term assessment given husk degradation and acidification (Kaboosi, Liaghat, and Hosseini 2012; Navabian, Ghane, and Hosseinzade 2017); aggregate sizing needs in-situ testing in clay soils (Byrne et al. 2023a); and synthetic geotextile envelopes must be evaluated individually because of varying physical properties, with filter-sock-plus-aggregate combinations not yet recommendable (Byrne et al. 2023b). Structural concrete alternatives—fly-ash- and coal-gangue-based concrete—remain untested for durability under high groundwater (J. Wang et al. 2016).

8.6 Phosphorus-removal media and filter design

Phosphorus-sorption materials (PSMs) constitute one of the most heavily populated gap areas, spanning selection, longevity, regeneration, and hydraulic integration. Z. Li and Arai (2020) argue that adsorbent choice must weigh phosphorus concentration, interfering ions, temperature, flow rate, and sediment load; prioritise selectivity at neutral pH; be testable for reuse and recovery; and be economical at field scale. Mendes et al. (2022) add that the apparently synergistic retention mechanism of filter media at high flow rates is unknown and that regeneration of spent media should be pursued. Iron- and steel-based media dominate specific proposals: optimising an iron-oxyhydroxide regeneration process and evaluating low-cost goethite such as bog iron ore (Allred, Martinez, and D. Gamble 2017); determining minimum contact time for zero-valent iron via higher-flow column and isotherm tests (Allred and Racharaks 2014); regenerating steel by-products by alkaline washing (Hua et al. 2016); connecting a steel-slag filter to a bioreactor only after the initial DOC flush and mixing fine steel to preserve conductivity (Sellner et al. 2019); optimising removal, retention time, slag mass, and cost (L. Wang et al. 2020); controlling sediment influx and improving flow distribution (Shedekar, Penn, et al. 2020); and recognising that ligand-exchange PSMs suit tile drainage only where structures allow high-frequency media replacement (every 4–6 months) (Penn, Livingston, et al. 2020). Fly-ash and pellet media raise saturation-timing questions—saturation time must be quantified before field use so pellets are replaced before shifting from phosphorus sink to source (S. Li, Cooke, L. Wang, et al. 2017; S. Li, Cooke, X. Huang, et al. 2018)—and filter blends remain unoptimised with respect to media configuration, contact time, and high-flow contact (Agrawal, K. King, Fischer, et al. 2011; Agrawal, K. King, Moore, et al. 2011; Kumari and Dong 2025; Kankarej and Upadhyaya 2023). Integrated woodchip-plus-PSM designs need multi-scale, multiyear evaluation of removal effectiveness, media placement, lifespan, and regeneration (Husk, Sanchez, et al. 2018).

Novel phosphorus dynamics are emerging. Z. Zhang et al. (2024) call for experiments on enhanced orthophosphate uptake by woodchips undergoing anoxic–oxic cycling and identification of polyphosphate-accumulating organisms, while redox-driven phosphorus release remains a monitoring gap requiring start-up flush and aging assessment with high-frequency dissolved-reactive-phosphorus (DRP) monitoring and drawdown plates (Bailon et al. 2025), multi-season confirmation of DRP removal (Rivas et al. 2020), more reactive passive-sampling resins (Z. Li, Chu, et al. 2020), and characterisation of dissolved organic phosphorus lability and bioavailability (J. Liu, Sævarsson, et al. 2024).

8.7 Aging, longevity, clogging, subsidence, and construction

The multi-decadal service life and physical stability of treatment structures are among the least-studied dimensions. Ghane, Feyereisen, Rosen, and Tschirner (2018) recommend longer study of woodchip particle-size reduction, investigation of whether woodchip carbon becomes recalcitrant to the point of inefficiency before particle-size reduction throttles flow, and studies beyond ten years. L. Christianson, Feyereisen, et al. (2020) similarly call for long-term documentation of biofilm formation, siltation, wood decay, and the effects of a decreasing carbon-to-nitrogen ratio and increasing lignocellulose index, to inform woodchip-recharge thresholds; long-term (>8 yr) field programmes were underway to establish efficacy (Greenan et al. 2009), and spatiotemporal removal should be evaluated across normal, wet, and drought years alongside woodchip C:N measurements (Hartfiel, Hoover, et al. 2023). Estimating replacement interval and cost depends on performance, lifespan, and residue cost, with pre-measured feedstock volumes proposed to control construction-method effects (Kohn et al. 2023). Chemical clogging is mechanistically unresolved and needs simulation models for prevention (C. Guo, J. Wu, et al. 2020; C. Guo, H. Yang, et al. 2021). Subsidence monitoring is a new methodological frontier: L. Christianson, R. Christianson, Hay, et al. (2025) propose applying a LiDAR method at additional sites—especially immediately after construction—to build subsidence time series and assess targeted inlet recharge. Sediment management at construction is itself a design frontier: Feyereisen, Ghane, et al. (2023) plan to weigh the opposing effects of bypass flow and nitrate removal in a sediment-exclusion system and advise pre-construction investigation of sedimentation sources. Network-scale diagnostics also need extension—higher-pressure air injection and time-domain electromagnetic resistivity monitoring for large drainage networks (Henine, Tournebize, et al. 2021).

8.8 Pumped, energy-assisted, and coupled configurations

Pumping to raise hydraulic loading is an emerging but incompletely validated avenue. Díaz-García et al. (2025) call for assessing different pumping configurations, note that continuous 24-h operation or a longer pumping window could increase treated flow, and caution that pumping may suit only the most appropriate applications. Hartfiel, Díaz-García, et al. (2024) advocate solar-powered pumped bioreactors to reduce operating and unit cost, and Hartfiel, Craig, and Soupir (2023) recommend sizing pumping to a bioreactor's actual rather than design capacity, timing pumping to source-water nitrate concentration, temperature, and carbon content, and extending applications to nitrate-laden surface waters, irrigation waters, ditches, cisterns, and groundwater. Management experiments—partial inlet-recharge strategies and dual-chamber operation engaging a second chamber only under high flow—are proposed (Hartfiel, Hoover, et al. 2023), and pumped constructed-wetland-plus-woodchip-bioreactor systems require analysis of particulate versus dissolved nutrient fractions during intense discharges and longer monitoring of the pumping effect on phosphorus retention (Pugliese, Skovgaard, et al. 2020). Novel source waters and enhanced configurations broaden

the space: denitrification responses to spring flow and concentration, with payment-based incentives for spring bioreactors (Easton, Bock, and Stephenson 2019); low-temperature bioelectrochemical reactors with horizontal flow, baffled oxidising/reducing zones, and plate cathodes (Law, Soupir, et al. 2018); and coupled phosphorus-removal structures sited at critical source areas, including a proposed redox-coupled Fe(II)–nitrate reaction for nitrate removal by activated aluminium (Penn, Williams, et al. 2024).

8.9 Wetlands, saturated buffers, and companion structures

Frontier questions extend to practices deployed alongside or instead of bioreactors. For saturated riparian buffers (SRBs), a key challenge is predicting nitrate removal well enough to design more effective buffers, motivating unsteady-flow and numerical-transport methods and optimal-width approximations (McEachran et al. 2020); replicating SRB results across landscapes, with biomass harvesting to sustain viability (Jaynes and Isenhart 2014); more sophisticated removal and hydrology components coupled to economic objective functions, and determination of maximum allowable buffer length before optimising width (Abdalaal and Ghane 2023); confirmation that removal increases as vegetation matures, with multipipe configurations and additional monitored sites (Jaynes and Isenhart 2019b; Jaynes and Isenhart 2019a; G. Singh et al. 2024); and paired-field determination of actual performance and the cost-effectiveness of the buffer-flow and controlled-drainage components (Ghane, Abdalaal, and Tehrani 2025).

Constructed and treatment wetlands present persistent performance-limit questions. Hourly-time-step dynamic modelling is needed to approximate nitrate removal in tile-fed wetlands with means to buffer and gradually release flows (Tanner and Sukias 2011), and long-term study is required of sustainable nitrogen-retention efficiency across climates, whether organic-carbon availability limits denitrification, the value of biomass cutting or slow-release carbon supplements, and flow-regime effects analysed through residence-time distributions (Tanner, M. Nguyen, and Sukias 2005; Tanner, M. Nguyen, and Sukias 2003; Tanner, M. Nguyen, and Sulkias 2005). A broad agenda for constructed treatment systems spans driving factors, flow–treatment relationships, design-parameter effects, long-term field tests, and winter-efficiency enhancement (Mendes 2021), alongside maturation of constructed wetlands for phosphorus retention (Mendes 2020), siting and sizing by wetland-to-catchment ratio and seasonal discharge (Steidl, Kalettka, and Bauwe 2019; Tournebize, Chaumont, and Mander 2017), and subtropical adaptation of land-treatment (FILTER) systems (Gardner et al. 2001).

Denitrification-bioreactor and combined-installation refinement remains open: reactor size, packing materials, elimination of preferential flow, and pH maintenance (Aide, Braden, and Svenson 2016); separating the contributions of controlled drainage with and without a bioreactor and evaluating clogging potential (Husk, B. Anderson, et al. 2017); and verification of design retention times and hydraulic conductivity (Feyereisen and L. Christianson 2015). Controlled and flow-regulating structures raise further questions—improving the connection of regulator elements to limit effective flow sizes (Popek et al. 2021), pipe-riser and event-scale trapping effects in water-and-sediment control basins (A. Gupta et al. 2019), and soil-surface clogging, storage geometry, and discharge-capacity effects in drainage-inlet flow control (Nédélec, Ouvry, et al. 2010). Finally, subsurface attenuation adjacent to drainage needs broader characterisation: hyporheic-zone processes and buffer widths sufficient to guarantee adequate residence times across hydrogeologic settings (Puckett and Hughes 2005), low-rate effluent irrigation to reduce direct transfers through mole-pipe drainage (Monaghan et al. 2007), and long-term monitoring of tillage and earthworm effects on phosphorus leaching (Ulén, Larsbo, et al. 2018). Collectively, these companion-practice frontiers reinforce the section's central message: moving edge-of-field treatment from site-scale demonstra-

tion toward validated, transferable design guidance capable of meeting watershed-scale goals across diverse climates, soils, and hydrologic regimes.

9 Salinity, Reclamation, and Drainage-Water Reuse in Arid and Irrigated Systems

Subsurface drainage is a cornerstone technology for reclaiming salt-affected land, yet its salt-leaching outcomes accrue slowly, are only partially observable with current programmes, and generate salt-laden water whose disposal remains unsolved. The frontier here is defined by this triad—slow and incompletely measured reclamation, the management of the water that reclamation produces, and the specialised chemistry of acid sulfate and irrigation-return systems—together with the sensing, modelling, and design tools needed to resolve it.

9.1 Salt-leaching dynamics and long-term reclamation trajectories

The recurring message is temporal: meaningful salt reduction often accrues so slowly that short observation windows systematically underestimate long-term benefit, and only extended drainage with sustained monitoring resolves the true trajectory (Almen et al. 2021; Daigh, Acharya, et al. 2025). The long-term effects of combined saline-water irrigation and subsurface drainage on both salinity and yield require further evaluation (G. Feng, Z. Zhang, C. Wan, et al. 2017; G. Feng, C. Zhu, et al. 2021), desalination rates and trends remain incompletely quantified (C. Guo, C. Yao, et al. 2024; C. Yao et al. 2024), and controlled drainage introduces unresolved temporal trade-offs, buffering outflow while risking salt accumulation (Jouni et al. 2018; P. Lu et al. 2019; Islam et al. 2025; Heng et al. 2021). Longer observation records paired with optimal leaching quotas and cost–benefit analysis are needed to close these gaps (H. Guo et al. 2025), and salt-tolerant crops in reclamation rotations, alongside quantification of rainfall and irrigation-efficiency gains relative to salt-removal capacity, remain to be characterised (Qadir, Ghafoor, and Murtaza 2001; Castanheira and Serralheiro 2010).

A parallel gap is measuring what actually leaves through the drains. Priorities include greater attention to surface-water monitoring of salt and fertiliser export (Mastrocicco et al. 2013), continuous flow meters and electrical-conductivity probes for oasis-scale balances with evaluation of impacts on downstream phreatic groundwater (Askri, Khodmi, and Bouhlila 2022), nutrient balances of recipient waters (Turtola and Paajanen 1995), and cluster wells with deep sampling to track salinity over time (A. Singh et al. 2020; Gafni and Zohar 2001). Process understanding of salt mobilisation is itself an active frontier: how macropores govern transient-state leaching and how their effects might be minimised (Xin, X. Yu, et al. 2016), the controlling factors in low-permeability and structured media (J. Zhang, Werner, and C. Lu 2022), salt replacement between pipes under brackish-water leaching (D. Li et al. 2023), rainfall–irrigation interactions and generalisable desalinisation ratios across climates and soils (S. Yu, J. Liu, et al. 2015; Y. Chen et al. 2025; X. Wang, Y. Zhang, et al. 2024), ponding strategies and leaching curves (RAO and LEEDSHARRISON 1991; Y. Yang, D. Li, et al. 2022), and the optimal root-zone matric potential that links leaching intensity to crop response (C. Zhang et al. 2021; Heng et al. 2021). Overarching these, waterlogging and salinity should be assessed together rather than separately (Valipour 2014), and the economic feasibility of drainage as a preventive measure against salinisation remains to be established (Datta, Jong, and O. Singh 2000; Halbac-Cotoara-Zamfir et al. 2022).

9.2 Drainage-water reuse, disposal, and treatment

Removing salt from soil relocates the problem to the drainage water, whose management is among the most persistent open challenges. Environmentally acceptable disposal methods are still needed (Ayars and Soppe 2014; Steppuhn 2006), evaporation-pond management must be tested at full scale and combined with volume- and contamination-reduction measures (BRADFORD et al. 1991), and trace-element fate—particularly arsenic speciation and uranium recovery or removal—compounds the disposal burden (Gao, Ryu, et al. 2007; Duff, Amrhein, and Bradford 1997; Gao, Tanji, et al. 2003). Emerging treatment routes remain at the demonstration stage and require testing under realistic conditions: electrolysis of saline wastewater at varying salinity (C. Yang, H. Liu, et al. 2023), sorptive filter materials evaluated with real rather than artificial drainage water (Pugliese, Canga, et al. 2023), and blending of drainage water with reverse-osmosis product water (A. Singh et al. 2020). Life-cycle cost assessment, scalable data infrastructure, and low-temperature crystallisation toward near-zero discharge define the longer-horizon disposal agenda (Qin et al. 2025).

Reuse research spans site-specific load–flow relationships, germination-stage salinity tolerance, and improved scheduling that meets crop evapotranspiration from a shallow water table while minimising upward salt flow (WESTCOT 1988). Additional needs include provisional field guidance for substituting drainage water on wheat (Sharma and Rao 1998), blending combined with evaporation-pond storage (Datta, De Jong, and O. Singh 2002), farm-level reuse embedded in a basin-wide salt balance (Qureshi et al. 2008), exploitation of seasonal groundwater undulation and drip conversion to prevent accumulation (Mirlas et al. 2021; Dou, Shi, R. Li, Miao, J. Yan, F. Tian, and B. Wang 2022; A. Singh 2019a), sequential reuse across cereals, grasses, forest plantations, and halophytes with innovative evaporation-pond designs (Sharma and Tyagi 2004), and land-treatment/pond combinations under restricted hydraulic loading with preferential-flow control and pre-treatment before wastewater application (Christen, Quayle, et al. 2010; Duan et al. 2015). Biochar, phytoremediation, and related reclamation innovations warrant further development (J. Liu, Xiong, et al. 2025; Steppuhn and McArthur 2017; Youngs and Leeds-Harrison 2000). Crucially, economic analysis indicates that feasible drainage-water recycling requires markedly higher and less volatile irrigated yields together with lower adoption costs than currently prevail (Sellars et al. 2022).

9.3 Acid sulfate soils and irrigation-return management

Drained acid sulfate (AS) and boreal soils raise a distinct set of specialised questions. Short-term hydro-meteorological controls on sulfide oxidation, ripening effects on hydraulic properties, and secondary-mineral formation should be incorporated into model applications (H. Salo, Virtanen, et al. 2023); the role of the nitrogen cycle in sulfide oxidation is unknown in boreal cultivated AS soils (Virtanen, Puustinen, and Yr-Hall 2017); the peat-cover thickness required to hinder oxidation of sulfidic material needs detailed study (Yli-Halla et al. 2022); and more detailed AS-soil maps, a clearer distinction between anthropogenic and organic acidity, and cessation of drainage of potential AS soils are urged (Åström et al. 2005). Amendment strategies—gypsum, elemental sulphur, and lime for long-term management of problem soils (MANN, PISSARRA, and VANHOORN 1982; Y. He et al. 2015), biochar selected for low contaminant content with urgent long-term impact assessment (Y. Tian et al. 2023), and optimum rates of organic conditioners that minimise adverse water-quality impacts (T. Zhang, Tan, Z. Zheng, Welacky, et al. 2015)—require longer-term verification.

For reclaimed saline, waterlogged, and Vertisol systems, priorities include how subsurface drainage integrated with crop and residue management affects soil quality, carbon sequestration, nutrient

budgets, and greenhouse-gas fluxes (Mukhopadhyay, Fagodiya, Prajapat, et al. 2023; Mukhopadhyay, Fagodiya, Narjary, et al. 2023), the long-term effects of controlled drainage on salinity (El-Ghannam et al. 2023), salt-crust formation and dissolved-species transport (Colombani et al. 2016), leaching effects on additional elements and organic modifiers (Soltani-Gerdefaramarzi et al. 2020), monitoring of elemental toxicities with waterlogging-stage-specific drainage scheduling (L. Qian et al. 2021), long-term root-growth resistance and drainable-porosity determination (I. Bahçeci, Nacar, et al. 2018; Bahceci 2008), additive-effectiveness and land-replacement characterisation (Isa et al. 2023), and multi-location, multi-condition testing over longer periods (Sadhukhan, Z. Qi, T. Zhang, Tan, L. Ma, and Andales 2019). On the irrigation-return side, needs span tolerant rootstocks, halophyte germplasm, and transgenic crops (Cuevas et al. 2019), land-use planning that steers irrigation toward low-salinity aquifers (Bethune, Q. Wang, and Gyies 2008), and sediment- and return-flow reduction from furrow-irrigated fields with improved pond removal and nitrogen-leaching-limiting management (Bjorneberg et al. 2020). Systematic evaluation of alternative field-scale irrigation and drainage strategies under ambiguity, expressed as a statistical likelihood of achieving goals, is advanced (Alzraiee, Gates, and Garcia 2013), supported by long-term geohydrological and crop data to validate simulations across irrigation/drainage systems (R. Yao et al. 2017). Retired land over shallow water tables, surrounded by irrigated fields, poses unresolved ultimate-use questions (Purkey and Wallender 2001).

9.4 Sensing, modelling, and design under salinity

Salinity-specific sensing is a fast-growing cluster: EMI mapping validated across soil orders and settings (Narjary et al. 2024), coupled apparent-conductivity–water-content models (S. Yu, Cheng, et al. 2025), and EM38 with multispectral and hyperspectral remote sensing integrated through interfaces that explicitly incorporate tile drains (A. Singh et al. 2020; Jayawardane et al. 2001). Coupled water–salt transport modelling remains immature. Models are needed that simulate salt and nitrate simultaneously in drainage water and the soil profile (Ebrahimian and Noory 2015); neural networks can predict effluent salinity from minimal input, but process codes such as SALTMOD may be preferred for detailed salt balances (Sarangi et al. 2006), and further evaluation must precede firm salt-transport recommendations (Hirekhan, S. Gupta, and Mishra 2007). Several strands press for tighter coupling of water and salt dynamics with unsaturated-zone models such as HYDRUS, SWAP, and UBMOD-S, incorporating regional layout parameters and multi-objective optimisation under climatic, textural, and groundwater variability (X. Han et al. 2025; G. Wang et al. 2025; Y. Wang, Sun, et al. 2024), and for root-growth modules, automatic TDR monitoring, and large-dataset validation (Y. Liu, Zeng, Ao, et al. 2022), extending to climate-change scenarios, deep-soil salt accumulation, and real-world scalability (Y. Liu, Ao, et al. 2026). Sustainability of saline-water irrigation warrants coupled HYDRUS–EPIC evaluation (G. Feng, Z. Zhang, and Z. Zhang 2019) and better representation of water-table contributions to storage and evapotranspiration (X. Jia, Scherer, et al. 2017). Earlier salinity series staged validation and salt-flow analysis across successive papers (KARAJEH, TANJI, and KING 1994), and proper representation of regional seepage would demand a fundamentally different modelling approach (Kelleners, Kamra, and Jhorar 2000).

Design under heterogeneity is the final gap. Integrated redesign of system, surface-water, and nutrient management is required (Eertwegh et al. 2006), together with texture-specific understanding of the economics of preferential subsurface drainage (Neha et al. 2022), and vertical porous flow pipes or fully charged segments to enhance salt drainage (X. Wang, Y. Zhang, et al. 2024; C. Yao et al. 2024; Jayawardane et al. 2001). New design criteria that protect crops while minimising both the salinity and the volume of drain water are sought (Christen, Ayars, and Hornbuckle 2001).

The most integrative vision unifies these threads: zonal design algorithms driven by a multi-scale water-and-salt transport model coupled to remote-sensing inversion for real-time early warning (Q. Zhu et al. 2025)—a target that ties salinity sensing, coupled modelling, and adaptive design into a single research programme.

10 Unintended Consequences and Multi-Objective Tradeoffs

Subsurface drainage and the reductive treatment systems appended to it are increasingly recognised for effects that fall outside the nitrate- and yield-centred accounting that has historically justified them. Removing a nutrient from one pathway can shift it to another, treating drainage water can manufacture new by-products, and altering the field water regime can set slow biogeochemical and pedological changes in motion. This section consolidates these cross-constituent consequences—greenhouse-gas coupling, pollution swapping, emerging contaminants, trace-element legacies, and drainage-induced soil evolution—as a single frontier defined less by settled magnitudes than by unmeasured tradeoffs. The recurring message is that few studies have quantified more than one consequence at a time, so the net balance among nitrate reduction, greenhouse-gas emission, secondary pollutant release, and long-term soil change remains largely conjectural.

10.1 Greenhouse-gas coupling and carbon export

Greenhouse-gas (GHG) coupling is a conspicuous gap in drainage accounting. Fluxes of carbon sequestration, pre-implementation emissions, and dissolved nitrous oxide in tile drainage and conservation-practice outflows need quantification (M. Mitchell, Newcomer-Johnson, et al. 2023), and coordinated large-scale studies are required because few have simultaneously measured N_2O, CH_4, and soil-organic-carbon change at the same sites (Hall et al. 2025). Longer-term dynamics of wetland carbon storage, N_2O, and CH_4, and the GHG effects of upgrading drainage infrastructure, remain unresolved (M. Mitchell, Shifflett, et al. 2022), and carbon budgets should be integrated into wetland-design models (Kadlec 2012).

A central strand couples nitrate leaching to indirect N_2O. Sawamoto et al. (2003) call for study of the nitrate–N_2O relationship in groundwater and of indirect N_2O from subsurface drainage, Minamikawa, Wagai, et al. (2015) argue for new calculation methods for indirect N_2O from leaching and runoff, and the tradeoff between indirect N_2O and heterotrophic-denitrification N removal must be evaluated across groundwater, drainage, river, and estuary (Minamikawa, Eguchi, et al. 2013; Minamikawa, Nishimura, et al. 2010). Open questions include nitrate–N_2O linkages when N application and precipitation are temporally disconnected (O'Brien et al. 2022), the magnitude and fate of N_2O losses (Burton, Wilts, and Macleod 2024), whether dissolved N_2O losses exceed surface emissions (E. Smith et al. 2019), and long-term cumulative dissolved-GHG effects of management (Oliveira, Foltz, et al. 2025). Webb, Clough, and Quayle (2021) outline an emissions-accounting agenda—seasonal emission-factor impacts, process-based modelling, gas-transfer-velocity models, the DOC/NO_3 ratio, and a global database—while Fontes et al. (2019) recommend year-round monitoring of both N_2O and nitrate leaching. Net-GHG evaluation in longer experiments (Preza-Fontes, L. Christianson, and Pittelkow 2023), the fate of waterborne methane alongside phosphorus balances when selecting measures (Carstensen et al. 2021), comparative life-cycle assessment from complete N balances (Hernandez-Ramirez et al. 2011), and biogeochemical measurement and modelling of methane production and oxidation in cold, poorly drained soils (Kumar, Nakajima, Kadono, et al. 2014) extend the frontier.

Dissolved-organic-carbon (DOC) dynamics and their ecological implications remain largely unknown (Ruark, Brouder, and Turco 2009; Furukawa, Shiratori, and Inubushi 2008), with dissolved-

organic-matter quality, microbial availability, and episodic drainage influence demanding paired-watershed designs and scale-appropriate sampling (Dalzell, J. King, et al. 2011; Bellmore et al. 2015). Treatment structures add their own GHG and carbon budgets: Brunton et al. (2024) recommend fuller assessment of full-size bioreactor methane dynamics, age effects on GHG consumption and release, "hot moment" production, and real-time control on outflow redox potential, and Strock, Ranaivoson, et al. (2025) note that a 2-h hydraulic residence time raises GHG emissions and that DOC export needs monitoring. Carbon-export accounting more broadly is a monitoring gap—aqueous total-carbon export for accurate sequestration accounting, multi-year data under variable climate, and the potential of water-level control to minimise dissolved-inorganic-carbon loss (Schilling, Streeter, Jones, et al. 2023).

10.2 Pollution swapping and by-product formation in treatment systems

Reductive treatment introduces tradeoffs that are only partly quantified. L. Christianson, Cooke, et al. (2021) name reducing pollution swapping—especially at start-up—and improving understanding of the removal of additional pollutants, compounds, and biologicals as key research areas. The by-products of concern span the reduction sequence. Nitrite accumulation should be measured in future bioreactor studies alongside long-term evaluation of phosphorus transport (Bell et al. 2015). Mercury methylation is a monitoring priority that Maxwell, Cooke, and L. Christianson (2022) suggest minimising through an outflow nitrate-concentration design criterion rather than a mass metric. Redox-driven phosphorus release is a recurring gap: Bailon et al. (2025) call for evaluating start-up flushes with immediate monitoring, aging bioreactors, and total phosphorus alongside higher-frequency dissolved reactive phosphorus (DRP) monitoring and drawdown plates, and Rivas et al. (2020) for continued multi-season monitoring to confirm longer-term DRP removal and the factors governing it. The mechanisms of enhanced phosphorus uptake by woodchips under anoxic–oxic cycling—including polyphosphate-accumulating organisms and whether anaerobic conditions are necessary—remain to be elucidated (Z. Zhang et al. 2024). Strock, Ranaivoson, et al. (2025) tie these together, calling for work on P-removal-material longevity, cold-condition optimisation, DOC-export monitoring, and strategies for managing P-treatment materials to minimise unintended phosphorus release, while Schilling, Streeter, Jones, et al. (2023) note that water-level control may curb dissolved-inorganic-carbon loss. Across these by-products the common prescription is the same: longer, higher-frequency, multi-constituent monitoring to bound tradeoffs that single-season demonstrations cannot resolve.

10.3 Emerging contaminants, pesticides, pathogens, and antibiotic resistance

Emerging contaminants and biological interactions broaden the constituent scope of drainage assessment. Priorities include better constraint of the mechanism and lifetime of glyphosate in soil and water (Spiese, M. Bowling, and Moeller 2023), planter modifications to reduce aquatic exposure to neonicotinoid and other seed-applied pesticides (Schaafsma et al. 2019), how watershed microbial communities respond to and control nutrient fluxes (Knorr, Weisener, and Phillips 2023), and compound-source identification across rainfall-runoff events (Fraindová et al. 2022). Particulate silicon transport coupled to warming and diatom dynamics (Sethna et al. 2022), nutrient stoichiometry and organic-matter fate in aquatic ecosystems (Manninen, Kanerva, et al. 2023; Manninen, Soinne, et al. 2018), and release of accumulated compounds and resistance genes in sludge-amended soils (Ghirardini and Verlicchi 2019) extend the list.

10.3.1 Contaminant transport through inlets and macropores

How contaminants enter and move through drainage networks is explicitly under-studied. S. Li, Bhattarai, et al. (2017) recommend research on how nitrate, phosphorus, and pesticides move through drainage inlets, and on inlet performance under crop-specific debris. The effective contributing area may differ for water, tracers, and reactive contaminants (Akay, Fox, and Simunek 2008), and controls on tile-sediment flocculation remain unresolved (Stone and Krishnappan 1997). Pesticide transport modelling forms a coherent cluster: particle-mediated colloidal algorithms and separate macropore accounting for leaching (Wendell et al. 2024), transport by mobile particles in convection–dispersion models (Zehe and Flühler 2001), a quasi-two-dimensional profile distribution and macropore–drain interconnectivity in chemical routing (Fox, Sabbagh, et al. 2007), slow-immobilisation and dissociation-kinetics processes for persistent pesticides (Novak, Banton, and Schiavon 2003), multi-season field monitoring across soil types (Dousset et al. 2004), and vulnerability functions from field characteristics (Leu et al. 2004). Analytical and marker methods—hydroxy-triazine determination (MUIR and BAKER 1978), a dicamba marker (Heaven et al. 2012), continued linuron monitoring (Gordon et al. 2005), and over-winter freezing/thawing effects on herbicide fate (Siimes et al. 2006) —complete this cluster.

10.3.2 Pathogens, microbial ecology, and antibiotic resistance

Microbial contaminant transport spans several open questions. Pathogen removal under lower residence times, *Salmonella* and *E. coli* removal mechanisms, and behaviour at higher flows need investigation (Soupir et al. 2018). The ecology of indicator bacteria in treatment wetlands—survival of faecal versus cryptic "*E. coli*-like" strains (Stott et al. 2023) and curli expression and downstream risk (Truhlar et al. 2015)—is unresolved, as is long-term pathogen leaching after manure application (Malik, Randall, and Goyal 2004). Fecal-source tracking needs corroborating markers (Gottschall et al. 2013), irrigation with wastewater warrants disinfection and reduced cold-season rates (Ding et al. 2021), and a broad catchment-scale microbial agenda spans riparian entrapment, constructed wetlands, and water as a human-infection vector (Collins et al. 2007). Antibiotic-resistance genes are an explicitly emerging frontier: their diversity, persistence, and dissemination require qPCR to complement metagenomics (S. Smith et al. 2019), and further study is needed to link resistance genes to host identity and transfer mechanisms (Rieke et al. 2018).

10.4 Trace-element legacies and ecotoxicology

Trace-element contamination associated with drainage in the western United States represents a long-horizon monitoring commitment. BENSON, DELAMORE, and HOFFMAN (1993) outline continued biological and groundwater monitoring, research into 10–20-year selenium-dissipation strategies, annual management planning, and pilot-scale microbial-volatilisation demonstration, and SAIKI and LOWE (1987) note further studies to determine whether aquatic animals elsewhere in the San Joaquin Valley are contaminated by selenium. Ecotoxicological monitoring of other contaminants extends the frontier: Campbell et al. (2023) call for mechanistic work on reduced amphibian cerebellum size, field examination of imidacloprid and its olefin metabolite, and regular monitoring of neonicotinoids near agricultural waterways with tighter restrictions to protect non-target organisms, while Dreyfuss, Vignoles, and Rondelaud (2018) note that effects on *Galba truncatula* raise conservation questions given its role as intermediate host of *Fasciola hepatica*. These constituents share a common feature: their consequences unfold over decades and reach organisms far removed from the field, so short monitoring windows systematically understate them.

10.5 Drainage-induced soil evolution, health, and biology

Slower-acting soil and biological responses to drainage are repeatedly flagged as understudied. The soil-biology response to changing drainage hydrology requires longer-term evaluation of biological indicators under standardised sampling (Sherbine et al. 2023), study of interactions between subsurface drainage and soil physicochemical properties and microbial activity (Y. Yang, Ao, et al. 2025), and continued research on microbial community structure and function to explain tillage and drainage impacts (Kaur, Nelson, Udawatta, et al. 2024; Kumar, Nakajima, Mbonimpa, et al. 2014). Robust soil-health diagnostics are sought—scalable nematode measurement within a soil-health index (N. Li et al. 2023), valid and repeatable soil-quality indicators in a systems framework (Nakajima, Lal, and S. Jiang 2015), soil-enzyme activities as management diagnostics (M. Singh et al. 2007), proof that no-till reduces nitrate leaching sustainably (Masri, Asher, and Piwarski 2024), and dynamic compaction indicators with long-term restoration trials (Caron et al. 2025). Specific biological questions include *Fusarium* abundance and soybean root rot under long-term drainage (G. Han et al. 2017), disruption of crack-to-burrow linkages by conservation tillage (Shipitalo, Nuutinen, and Butt 2004), evapotranspiration quantification to explain drain-flow differences (Daigh, X. Zhou, et al. 2014), cultivar drought tolerance guiding system selection (Darzi-Naftchali et al. 2017), and forage-sorghum nitrate testing (Nash et al. 2017).

A distinctive line treats subsurface drainage as a driver of long-term, potentially chaotic soil evolution. Montagne, Cornu, et al. (2009) argue that quantifying human-induced soil evolution as a function of time—including drainage's impact on clay-mineral dissolution—should be a research priority, and that soil evolution under climate change must be explicitly studied and incorporated in global models (Montagne and Cornu 2010); significant material losses over a few decades make anthropogenic perturbation a powerful tool for studying soil-evolution timescales (Montagne, Cousin, et al. 2013). Whether drainage-induced evolution continues or stabilises over decades, resolved through measurements along drains at different slope positions and network ages, remains open (Frison et al. 2009), and a long classical experiment continues to test whether soil structure is eventually affected (HOOGHOUDT 1952). Empirical prediction of earthworm activity from tillage, residue cover, and particle size is proposed (Ouellet et al. 2008), and sustained drainage operation and cultivation is framed as a management imperative for long-term soil health (M. Singh et al. 2007). Collectively, these consequences reframe drainage as an agent of slow, coupled biogeochemical and pedological change whose full cost cannot be read from a few years of outlet monitoring.

11 Toward Integrated, Targeted, and Adoptable Solutions

The preceding frontiers make clear that better process understanding, models, and monitoring are necessary but not sufficient: translating frontier science into water-quality and productivity gains depends on how practices are combined, where they are placed, how they perform under a changing climate, and whether they are economically and institutionally adoptable. This section draws the integrative and socio-economic strands of the literature together. Its recurring message is that no single practice suffices, that biophysical questions are inseparable from human ones, and that the decisive advances will come from coupling mechanistic understanding with spatial targeting, economics, policy, and shared decision-support infrastructure.

11.1 Bundling, sequencing, and spatial targeting of practices

A pervasive conclusion is that no single best management practice (BMP) is a silver bullet (Sunohara, Gottschall, Craiovan, et al. 2016), and that multiple BMPs in series are needed—with in-

field/edge-of-field combinations judged highly promising yet little studied, and wetland and water-retention measures still to be developed for sensitive downstream waters (Lenhart et al. 2016). Attaining water-quality goals will require combinations implemented at a high level in a targeted manner (L. Christianson, Knoot, et al. 2014). Open questions concern how best to pair conservation tillage with cover crops and in-field with edge-of-field practices while accounting for legacy effects (Makhtoumi, Aragon, and Lark 2025), how to formalise optimisation of synergistic combinations (Kreig et al. 2019), and how practices interact and what nitrogen-removal capacity each provides given better mapping of tile-drained land (McLellan et al. 2015). Broad programmes are proposed to compare drained and undrained systems, to bundle bioreactors, phosphorus reactors, controlled drainage (CD), and low-impact development alongside internal-loading management (Baulch et al. 2019; Husk, Julian, et al. 2024; Husk, B. Anderson, et al. 2017), to develop cost-effective BMPs that support cover crops and mitigate warming (R. Gupta, Bhattarai, Dokoohaki, et al. 2023), and to test intercrops and hydrology-altering management for phosphorus reduction (Hernandez-Ramirez et al. 2011; GAYNOR and FINDLAY 1995).

Closely related is targeting practices to the parts of the landscape that matter most. Identifying hydrologically active runoff-generating areas as mitigation priorities is recommended (Melland et al. 2008), but whether elevated loads originate from many diffuse sources or a few hot spots (Irvine et al. 2019), whether targeted or regionally saturated cover-crop implementation is needed for consistent reductions (Speir et al. 2022), and how to delineate zones of higher nitrate leaching (Bakhsh and Kanwar 2005) remain open. Targeting critical source areas and hydrologically connected hotspots is repeatedly stressed (McDowell 2015; Brendel et al. 2019), prioritisation is expected to improve the return on conservation investment (Makhtoumi, Aragon, and Lark 2025), and automating spatial-targeting principles into an accessible GIS tool with whole-basin surveys and cost–benefit analysis is a priority (Kvítek et al. 2023). Further scaling needs include combining GIS with hydrological modelling and saturation risk to find nitrate thresholds across soil texture and temperature (G. Smith et al. 2023), resolving how drainage density controls stream solute dynamics (Kennedy et al. 2012), and identifying critical precipitation periods rather than annual averages (T. Salo and Turtola 2006).

11.2 Climate change, extremes, and system resilience

Climate adaptation frames many priorities. The overarching need is to keep identifying practices and combinations that increase system resilience to climate extremes across additional soils and longer periods (Saha et al. 2024), supported by modelling agendas spanning additional periods, representative concentration pathways, management options, cultivars, cover-crop types, and soils (Malone, Garbrecht, et al. 2020), and by field-scale integrated biophysical modelling that resolves within-county variation in fertiliser and manure inputs under multiple CMIP5/CMIP6 forcings with attention to perennial-grass co-benefits (Ferin et al. 2023). A central design challenge is cropping and drainage systems that function under both drought and flood, including engineering solutions to capture excess floodwater (Dietzel et al. 2016). Adapting drainage design itself to a changing climate is an explicit priority, tied to understanding weather–soil–groundwater–discharge interactions under increased precipitation and to evaluating designs over longer periods and different climates (Deelstra 2015; Tuohy, O'Loughlin, and Fenton 2018; Malota, Taguta, et al. 2024; Deelstra, Iital, et al. 2014a; Deelstra, Iital, et al. 2014b). Remaining needs span the costs, benefits, and payback of adaptation and carbon-farming knowledge exchange (Peltonen-Sainio, Sorvali, and Kaseva 2021), the effect of extreme events on cover-crop performance (R. Gupta, Bhattarai, Kalita, et al. 2023), nitrogen consequences under warming (Haghnazari et al. 2020), and testing across additional rainfall environments, soils, and rotations to confirm drainage–management interactions (Takeshima et al.

2023; Y. Liu, Zeng, Y. Hu, et al. 2025; Grewal et al. 2025; Negm, Youssef, Skaggs, et al. 2014; BENOIT, FISHER, and BORNSTEIN 1967).

11.3 Economics, incentives, and cost-effectiveness

A large share of the frontier is economic. Cost–benefit and cost-effectiveness analyses are repeatedly flagged as future work—for surface amendments and energy savings (Wamono et al. 2016), comprehensive short- and long-term subsurface-drainage benefits (H. Guo et al. 2025), paddy-field cost and consumption (Z. Jia et al. 2023), edge-of-field cost-efficiency (Mendes 2020), and the opportunity cost of drainage investment together with policies that motivate improvement (Wichelns 2002; THOMAS et al. 1995; Kladivko, Willoughby, and Santini 2005; Saadat, L. Bowling, Frankenberger, and Brooks 2017; Manik et al. 2019; Helmers et al. 2012). Identifying long-term savings—such as greater fertiliser-nitrogen-use efficiency—to offset higher installation costs, alongside incentive and cost-share design, is recommended (Castellano et al. 2019; Sunohara, Gottschall, Wilkes, et al. 2015), and cost reduction is itself a research target through local manufacture and new low-cost techniques and materials (Datta, De Jong, and O. Singh 2002; Yannopoulos et al. 2020). Complementary needs include advanced case-by-case cost analyses to promote 4R implementation (W. Liu, Yuan, and Koropeckyj-Cox 2021), more data on the cost-effectiveness and ecological benefits of conservation practices (Yuan et al. 2022), and explicit incorporation of practice costs into future modelling (McLellan et al. 2015). Incentive mechanisms warrant study: continued positive incentives and greater weight on non-excludable benefits in multifunctional landscapes (L. Christianson, Knoot, et al. 2014), cost-share and risk-insurance instruments (Jaynes 2012; Karlen et al. 2005), and nutrient-trading programmes for shallow-drain adoption (Ghane and Askar 2021). Feasibility analyses caution that drainage-water recycling requires markedly higher, less volatile irrigated yields and lower adoption costs (Sellars et al. 2022), motivating work on the economics of cover crops (Waring, Lagzdins, et al. 2020), long-term cumulative yield and profit of drainage installation (Mourtzinis et al. 2021), and integrated biophysical–economic modelling at field scale (Ferin et al. 2023).

11.4 Policy, institutions, adoption, and the human dimension

Biophysical understanding alone is insufficient; governance, capacity, and farmer engagement are decisive. A repeated recommendation is to shift from standardised toward flexible, location-specific solutions that reflect stakeholder preferences, balancing top-down against bottom-up participatory management and linking farmers' tacit knowledge with researchers' explicit knowledge (Ritzema and Schultz 2011; Ritzema 2016; Ritzema, Abdel-Dayem, et al. 2023). Capacity development, education, and policy change to lower adoption barriers are emphasised (Castellano et al. 2019; SHIRMOHAMMADI, WENBERG, and RITTER 1995; Kesicka et al. 2023), along with governance frameworks, inter-agency coordination, dedicated drainage institutions, and time-bound action plans (A. Singh 2019c; Qureshi et al. 2008; Gopalakrishnan and Kulkarni 2007; Ritzema, Satyanarayana, et al. 2008; Valipour 2014; Quinn 2009), and awareness, subsidy, micro-finance, and participatory operation-and-maintenance programmes (Satyanarayana and Boonstra 2007; Manjunatha et al. 2004; Ben Aissa et al. 2013; MCFARLANE and COX 1992).

Adoption and social dimensions are stressed across contexts. Detailed cost–benefit analyses, joint farmer–researcher experimentation, and capacity building are prioritised, with adoption ultimately depending on farmers' long-term cost–benefit perspective (Okwany et al. 2016); creating know-how and farmer confidence before project execution is essential (Ghumman et al. 2011); and key practice-use issues—where to apply BMPs, how to bundle them, and transferability to

commercial farms—remain open (Ward et al. 2018). Stakeholder willingness to adopt must inform proposed measures (Cuevas et al. 2019), research integrating ecological, pollution-related, and economic impacts with models for optimal policy design is urged (Saikkonen et al. 2014), and incentivising on-farm research while scaling individual efforts to meso- and macro-scale policy is called for (David, Flint, et al. 2015). Policy reforms sought include legislative reassessment to remove barriers to natural-water-retention measures (Kvítek et al. 2023), national support to retain open ditch-and-field networks (Marja et al. 2013), producer-participation and pilot-programme information for policy optimisation (Malone, Radke, et al. 2023), and integrated packages—reduced plantings, farmer compensation, and gradual subsidy liberalisation—in salinity contexts (Kotb et al. 2000).

11.5 Integrative landscape-scale frameworks and decision support

The most explicit long-range vision is numerically explicit landscape-scale models that couple the processes governing ditch services, optimise the spatial arrangement of practices according to connectivity, and define multi-service maximisation strategies informed by stakeholder-valued rankings—prerequisite to which is better mapping of ditch hydraulic, vegetation, and sorption properties (Dollinger et al. 2015; Herzon and Helenius 2008). Landscape-scale approaches that combine on-farm change with off-site wetland and riparian restoration are broadly endorsed (Blann et al. 2009; Goeller et al. 2019), and truly integrated watershed management requires better information on controlling processes such as the groundwater-head distribution linking topography and geology to the drain fraction (Mahmood et al. 2023). Coupled field–watershed models enable water-table-management scenario analyses not possible with component models used separately (Dayyani et al. 2010), frameworks should remain extensible to additional processes and dynamically calibrated indices (Basu et al. 2010; Tao, J. Liu, Guan, et al. 2025; A. Verma, S. Gupta, and Isaac 2014), and modelling should increasingly guide practice placement and stacking (Kröger et al. 2013; Tomer et al. 2015; Lemke et al. 2011; McDowell, Monaghan, et al. 2021; K. King, Hanrahan, et al. 2022; Roley et al. 2016).

Emerging sensing and data technologies are widely seen as enabling this next phase. Broad agendas span optimal conditions for detecting drainage patterns and extracting drain lines, finer-resolution soil moisture, machine-learning mapping of soil properties, and within-field nitrogen-stress mapping (Khanal et al. 2020), alongside spatial frameworks that combine climate and soil type with geo-referenced farmer-reported data to estimate field-specific benefits (Cassman et al. 2025). Decision-support needs include validating aggregate edge-of-field losses and delivery ratios (K. Reid, K. Schneider, and McConkey 2018), testing soil-wetness-index tools across soils (Julien et al. 2023), integrating water and nutrient management with precision agriculture (Jung et al. 2024), and coupling downscaled climate information with an understanding of how farmers incorporate excess-precipitation experience into decisions (Morton et al. 2015). Better indices and shared tools are sought—phosphorus and nutrient indices that represent subsurface drainage, wetland and riparian-buffer removal, and stream-network connectivity (Buczko and Kuchenbuch 2007; Pärn, Pinay, and Mander 2012), extensible nutrient-credit-trading tools for drainage water management (Negm, Youssef, Chescheir, et al. 2016), a Soil Vulnerability Index that distinguishes surface from subsurface drainage (Baffaut et al. 2020), and coupling with ecosystem-service tools such as INVEST to assess tradeoffs (McLellan et al. 2015). Underlying all of these is the data imperative: better, longer, and more spatially representative datasets—spanning water and solute fluxes, greenhouse-gas emissions, root and biomass development, and management history—are the foundational requirement for validating models, resolving tradeoffs, and delivering site-specific, climate-resilient guidance (Z. Ma et al. 2025; SKAGGS, BREVE, and GILLIAM 1994; Deng et al. 2011).

12 Conclusions

Read as a whole, the frontier of subsurface-drainage research is defined less by an absence of ideas than by an absence of the evidence, tools, and coordination needed to convert ideas into transferable guidance. Across every domain surveyed here—monitoring, modelling, network mapping, nutrient and water-table management, edge-of-field treatment, salinity and reuse, unintended consequences, and adoption—the field's most secure statements are about the limits of its own evidence. Headline results rest disproportionately on short records, single sites, and narrow agro-ecological windows, and their authors caution explicitly against premature generalisation. The recurring signal is therefore a precondition rather than a conclusion: progress in each domain is bounded by the length, density, spatial replication, and consistency of the underlying observations (Rogger et al. 2017; Kladivko, Brown, and J. Baker 2001). This makes the observational foundation a frontier in its own right, and one on which all others depend.

Four converging priorities emerge from that shared diagnosis. The first is coordinated, long-term, multi-scale data. The single most widely voiced need is for records long enough to separate management effects from climatic noise and from slow, directional change in soils and hydrology, dense and year-round enough to resolve the non-growing-season and event-scale windows in which losses concentrate, and geographically broad enough to test whether site-specific findings hold across other soils, climates, crops, and management regimes. Layered onto record length are harmonisation and comparability: standardised methods, complete site characterisation, shared databases, and openly reported campaigns, without which accumulating studies cannot support the meta-analyses and typologies that would generalise them. Tracers, isotopes, and high-frequency source apportionment recur as the means to resolve the pathways that bulk flow-and-concentration records leave ambiguous, and experimental designs capable of detecting the effect sought—paired basins, nested scales, and defensible analysis of the unreplicated field trials that dominate the literature—form the bridge between a lengthening record and transferable conclusions.

The second priority is strengthened and validated model components. Because field trials cannot span every combination of practice, soil, and climate, process-based drainage and water-quality models are expected to carry much of the predictive burden; yet their credibility is constrained less by their formulation than by the narrowness of the conditions under which they have been tested. Validation across a wider envelope, the replacement of empirical coefficients with functions of measurable site attributes, and the process-level measurements needed to constrain the least mature routines— denitrification and the nitrogen cycle, phosphorus and its dominant preferential-flow pathway, cold-season dynamics, and coupled seepage and groundwater exchange—are near-universal priorities (Youssef, Skaggs, et al. 2006; Radcliffe et al. 2015). So too are explicit attention to identifiability and uncertainty, so that models are reported with propagated bounds rather than as point predictions, and the disciplined integration of machine learning and coupled model structures, whose own validation requirements the field is only beginning to confront. A parallel and equally unresolved gap is spatial: the physical drainage network is largely unmapped, so uncertainty about the network itself propagates into every downstream prediction, and translating instrumented-plot knowledge to the watersheds where water-quality goals are actually set remains a genuinely open methodological problem requiring non-invasive detection, shared databases, intensively monitored bridging catchments, and mechanistic treatment of channel-network, connectivity, and depressional processes (Amado et al. 2018; Naz, Ale, and L. Bowling 2009).

The third priority is bundled, targeted, and climate-resilient management. Across nutrients, water-table control, drainage design, and edge-of-field treatment, the levers are better described than optimised. No single practice is a silver bullet; the decisive gains lie in combining in-field and edge-of-field practices, sequencing them, and placing them where they matter most—on the

hydrologically active areas, critical source areas, and connected hotspots that generate a disproportionate share of loads. The common thread is a shift from passive, uniform prescription toward active, location-specific control: from one-way conveyance toward managed water tables, automation, and forecast-based operation; from agronomic nutrient rates toward site-specific crediting of legacy and mineralisable sources and management targeted to the temporal windows of loss; from demonstration-scale treatment structures toward standardised, transferable design across climates and maturation stages (Randall and Mulla 2001; Strock, Sands, and Helmers 2011; L. Christianson, Cooke, et al. 2021). Underlying all of these is climate adaptation: cropping and drainage systems, and the design criteria that govern them, must function under both drought and flood, and must be re-evaluated against long records rather than the short-term simulations on which much current guidance rests.

The fourth priority is interdisciplinary work that links mechanism, tools, and viable adoption, and it is inseparable from a growing recognition of consequences that fall outside the nitrate- and yield-centred accounting that has historically justified drainage. Removing a nutrient from one pathway can shift it to another; treating drainage water can manufacture new by-products; and altering the field water regime can set slow biogeochemical and pedological change in motion. Greenhouse-gas coupling, pollution swapping, emerging contaminants, trace-element legacies, and drainage-induced soil evolution share a common feature: few studies have quantified more than one consequence at a time, so the net balance among competing objectives remains largely conjectural, and their full costs cannot be read from a few years of outlet monitoring (M. Mitchell, Newcomer-Johnson, et al. 2023; Kadlec 2012). Resolving these tradeoffs, and translating frontier science into practice, ultimately depends on the human dimension—economics, incentives, policy, institutions, and farmer engagement—and on integrative landscape-scale frameworks and shared decision-support infrastructure that couple mechanistic understanding with spatial targeting, cost-effectiveness, and stakeholder knowledge (McLellan et al. 2015; L. Christianson, Knoot, et al. 2014). Biophysical insight alone is necessary but not sufficient; without economically and institutionally adoptable solutions, better science does not become better outcomes.

These four strands are mutually reinforcing rather than independent. Longer, harmonised data are what allow model components to be validated and their uncertainty bounded; validated, coupled models are what allow field-scale knowledge to be scaled to watersheds and management to be targeted and made climate-resilient; and targeted, multi-objective management is what makes the consequences worth quantifying and the economics worth analysing. The unifying message of this review is therefore that the next decade of progress in subsurface drainage will be secured less by new equations or novel practices than by the sustained, well-instrumented, openly reported, and geographically diverse field campaigns needed to test the ideas already in hand, by the coordinated data infrastructure and standards that make those campaigns cumulative, and by the interdisciplinary integration that ties mechanism to tools and tools to adoption. The task ahead is one of consolidation and coordination at scale—an agenda in which data, models, management, and human-dimension advances must progress together, because each is bounded by the others.

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